Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

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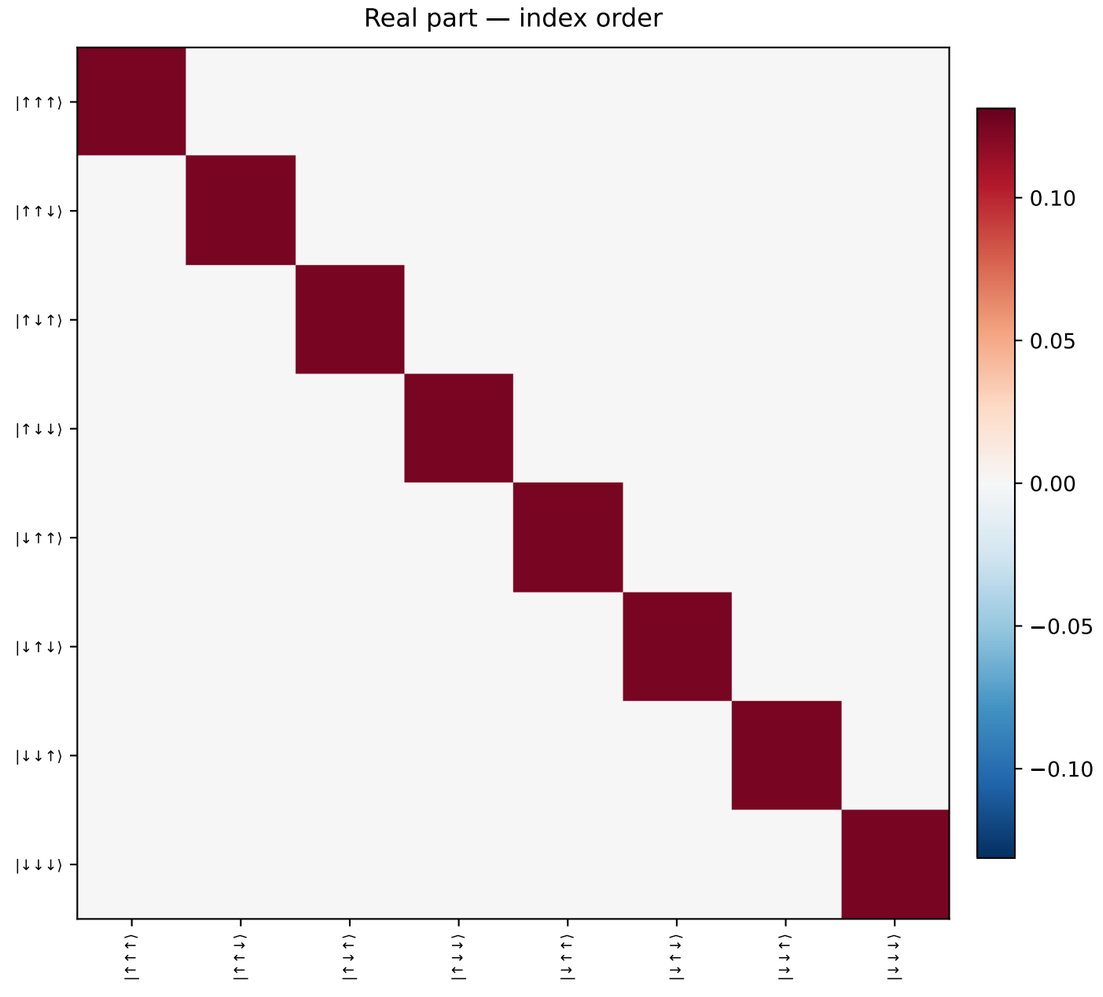
Body order : 3
Clusters   : 41

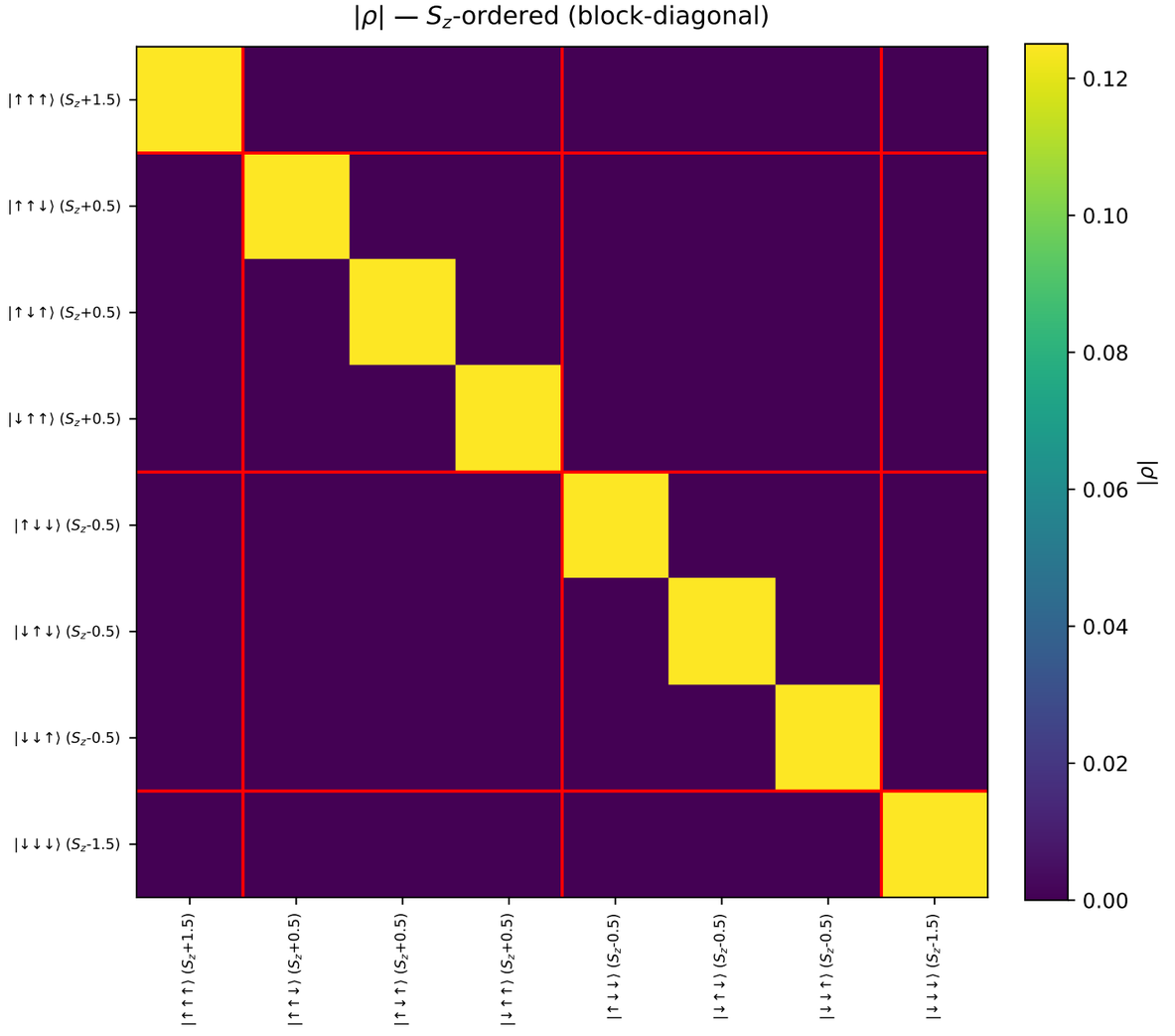
Date: 2026-07-11 15:46

Summary — 3-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1,2	0.125000	—	0.000000	0.000319	—	0.000000	STRONGLY MIXED
0,1,3	0.148891	—	0.000000	0.002603	—	0.000000	STRONGLY MIXED
0,1,4	0.125559	—	0.000000	0.000333	—	0.405751	STRONGLY MIXED
0,1,5	0.125067	—	0.000000	0.001944	—	0.417046	STRONGLY MIXED
0,2,3	0.125559	—	0.000000	0.000337	—	0.000000	STRONGLY MIXED
0,2,4	0.148904	—	0.000000	0.000525	—	0.000000	STRONGLY MIXED
0,2,5	0.125065	—	0.000000	0.000814	—	0.417074	STRONGLY MIXED
0,3,4	0.125949	—	0.000000	0.000001	—	0.000000	STRONGLY MIXED
0,3,5	0.150430	—	0.000000	0.000075	—	0.315200	STRONGLY MIXED
0,4,5	0.150975	—	0.000000	0.000379	—	0.314028	STRONGLY MIXED
1,2,3	0.496207	—	0.000000	0.028891	—	0.088963	MIXED
1,2,4	0.486887	—	0.000000	0.002873	—	0.088517	MIXED
1,2,5	0.125000	—	0.000000	0.002806	—	0.422828	STRONGLY MIXED
1,3,4	0.499664	—	0.000000	0.070670	—	0.087949	MIXED
1,3,5	0.145172	—	0.000000	0.000015	—	0.350783	STRONGLY MIXED
1,4,5	0.125941	—	0.000000	0.000066	—	0.445233	STRONGLY MIXED
2,3,4	0.499782	—	0.000000	0.073909	—	0.087901	MIXED
2,3,5	0.125967	—	0.000000	0.000218	—	0.445647	STRONGLY MIXED
2,4,5	0.144255	—	0.000000	0.000665	—	0.353025	STRONGLY MIXED
3,4,5	0.125368	—	0.000000	0.000010	—	0.413806	STRONGLY MIXED

3-Body RDM (Sites 0,1,2)





Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1250000348
  Von Neumann entropy  : 2.0794414025
  Eigenvalue min       : 1.248858e-01
  Eigenvalue max       : 1.250381e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12500003

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

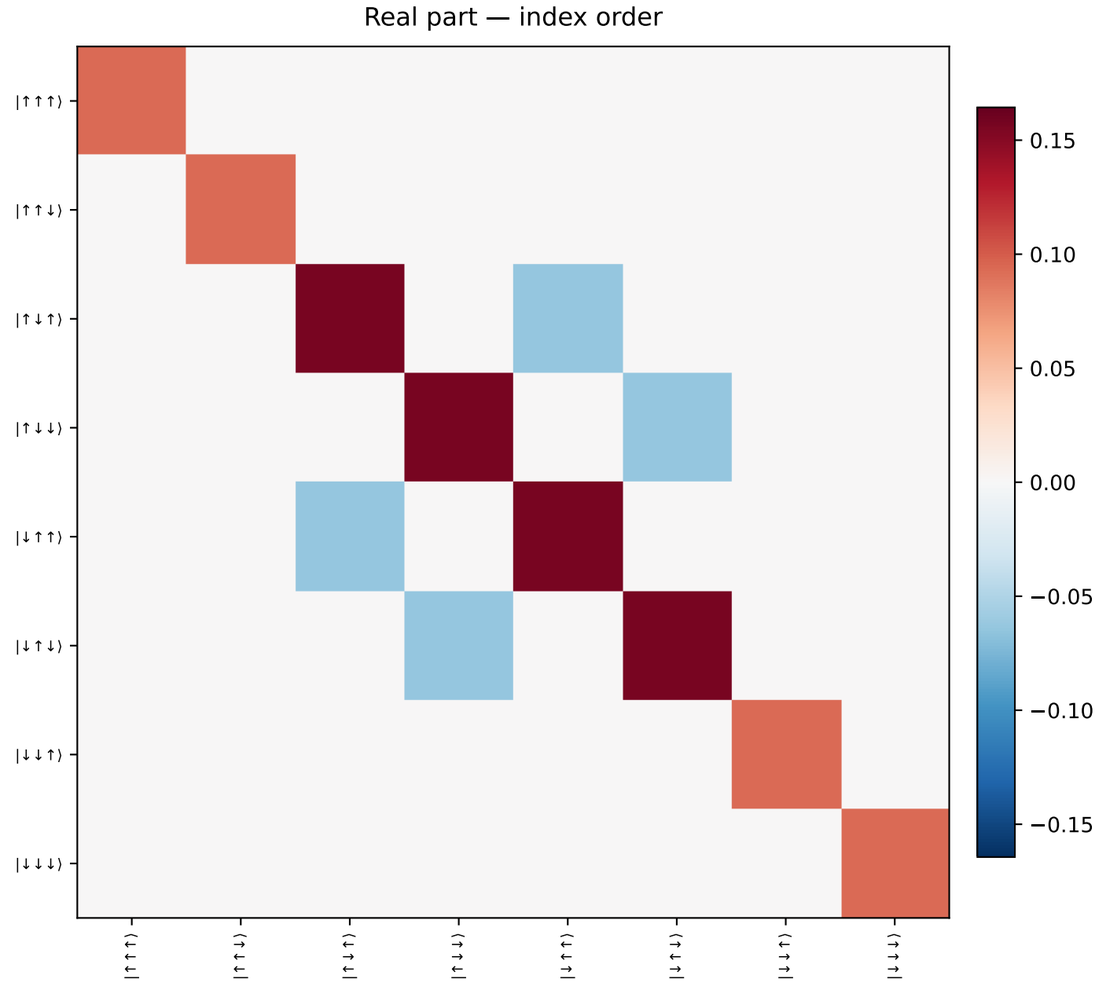
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000001

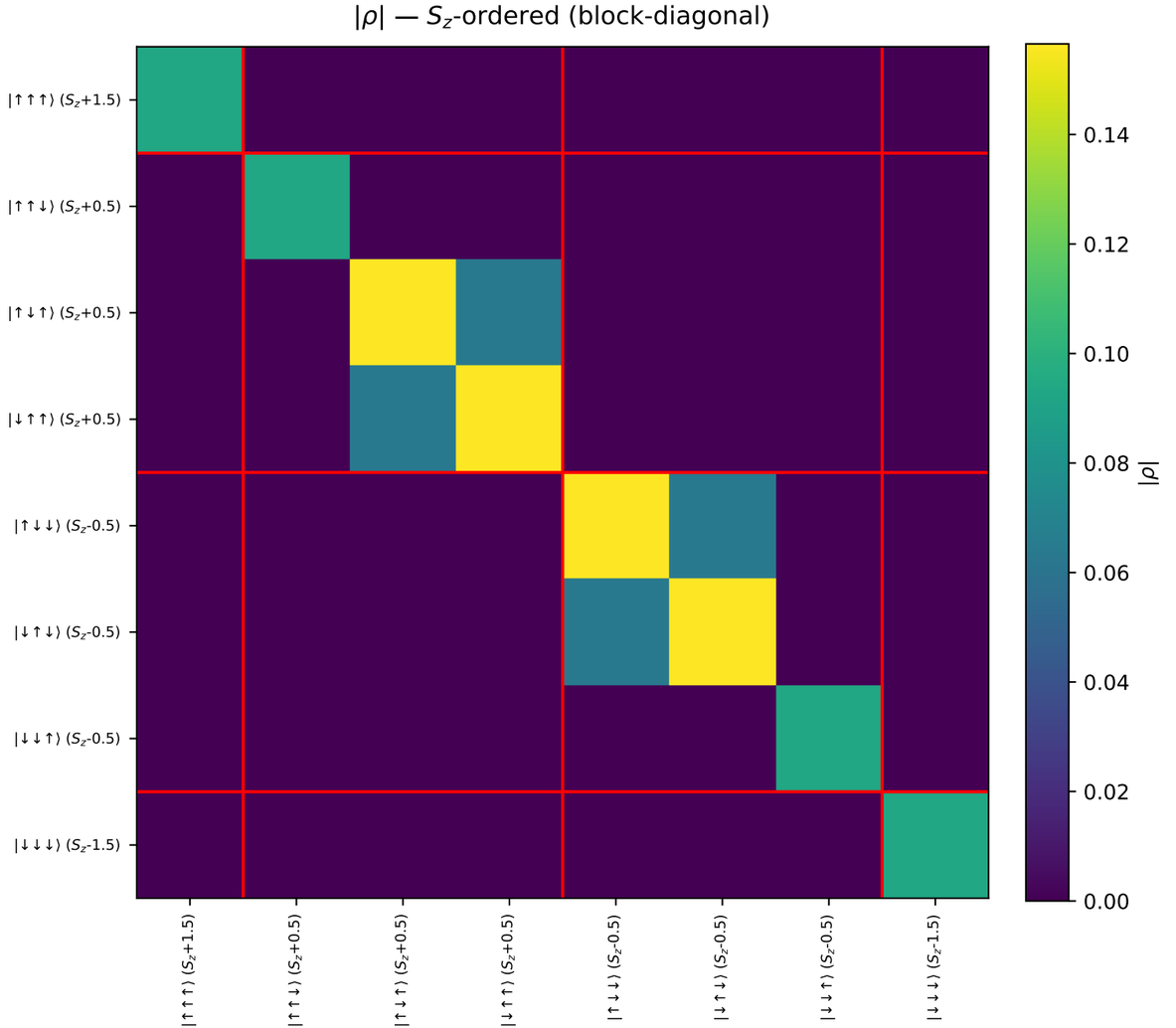
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.105606  C_GME=0.000015
2. p=0.068840  C_GME=0.000056
3. p=0.117924  C_GME=0.000135
4. p=0.115502  C_GME=0.001845
5. p=0.091682  C_GME=0.000017
... and 5 more states
```

3-Body RDM (Sites 0,1,3)





Trace=1.000000 Purity=0.148891 Entropy=1.994894 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,1,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1488911426
  Von Neumann entropy  : 1.9948940631
  Eigenvalue min       : 9.344902e-02
  Eigenvalue max       : 2.196529e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.14889114

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

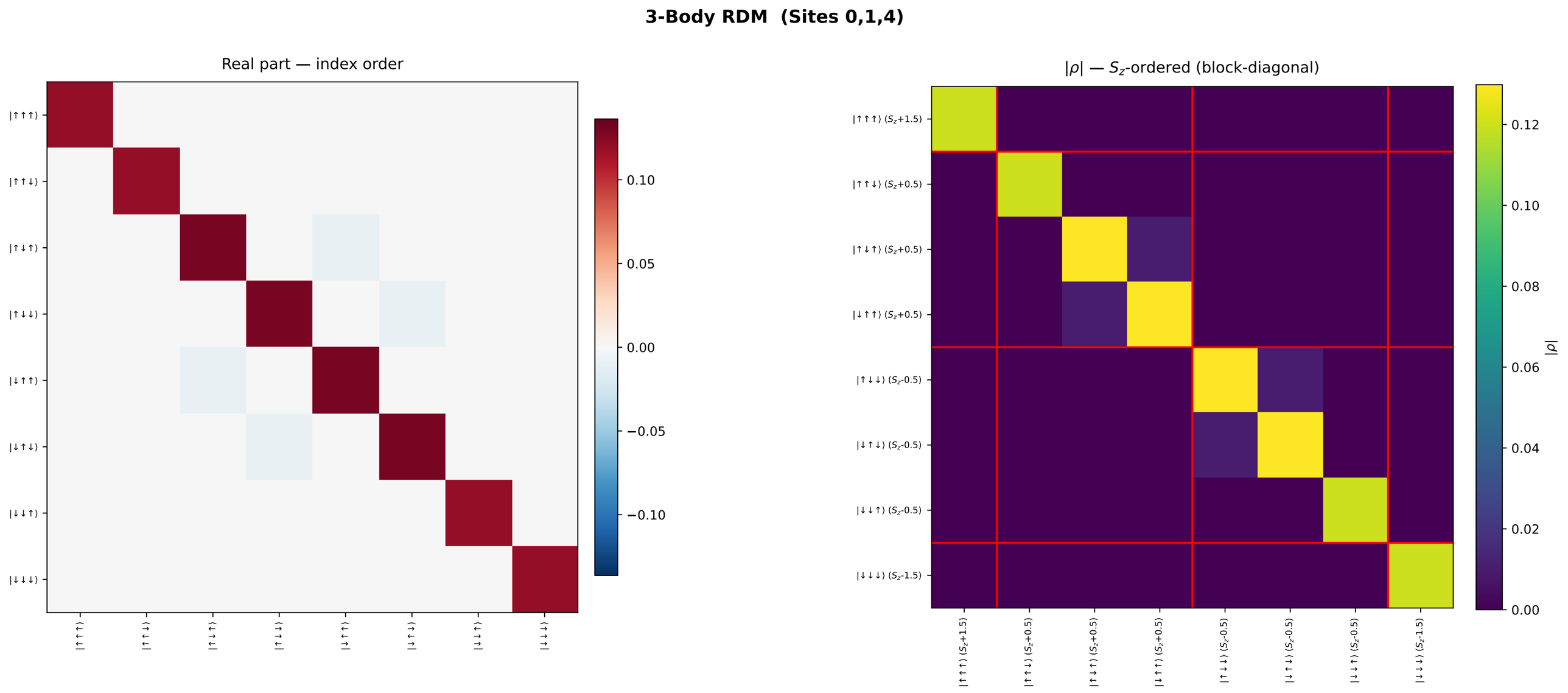
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000000

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.102787  C_GME=0.000000
2. p=0.099340  C_GME=0.000002
3. p=0.084777  C_GME=0.000000
4. p=0.130842  C_GME=0.013100
5. p=0.088127  C_GME=0.009901
... and 5 more states
```


3-Body RDM (Sites 0,1,4)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125559 Entropy=2.077259 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

Analysis Summary - 3-Body RDM (Sites 0,1,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1255590621
  Von Neumann entropy  : 2.0772592054
  Eigenvalue min       : 1.201735e-01
  Eigenvalue max       : 1.394792e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12555906

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

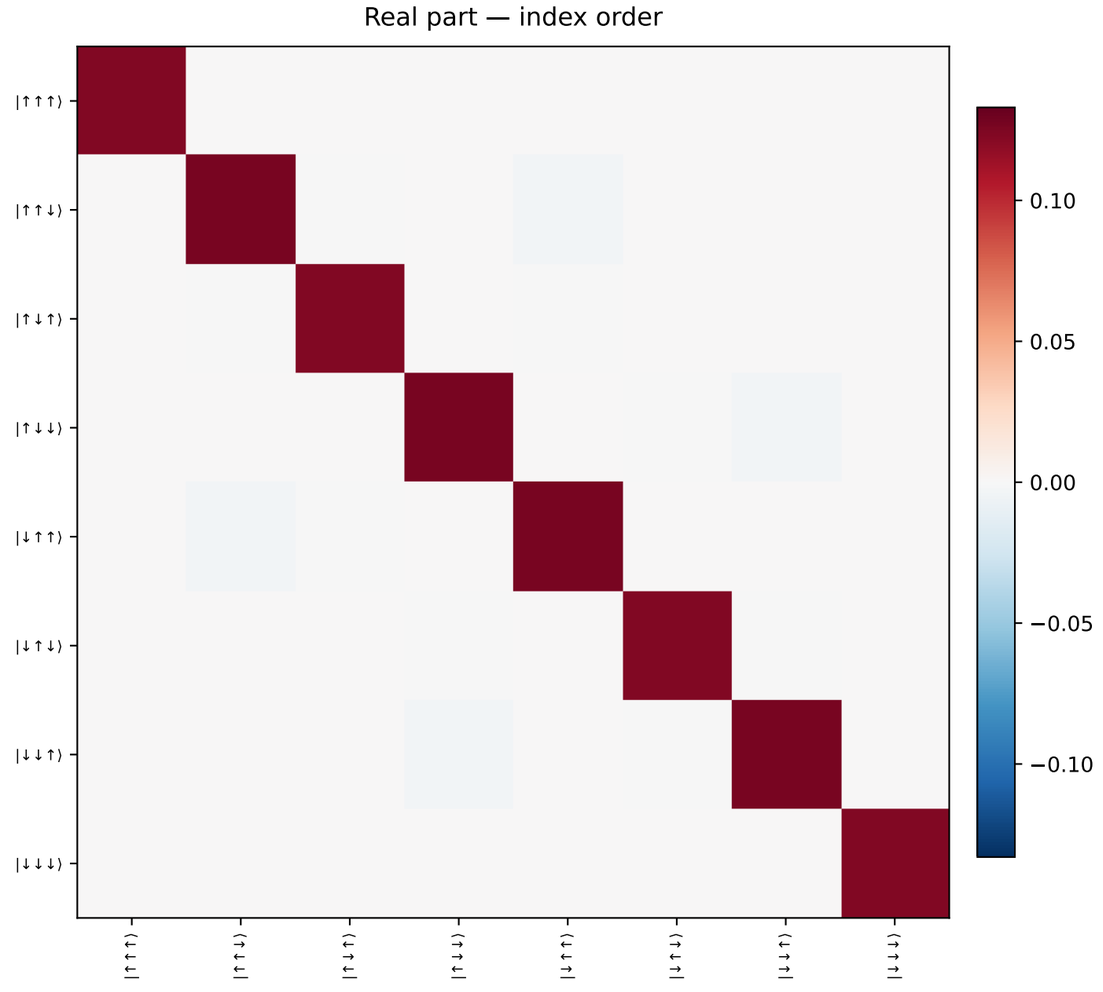
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.40575069

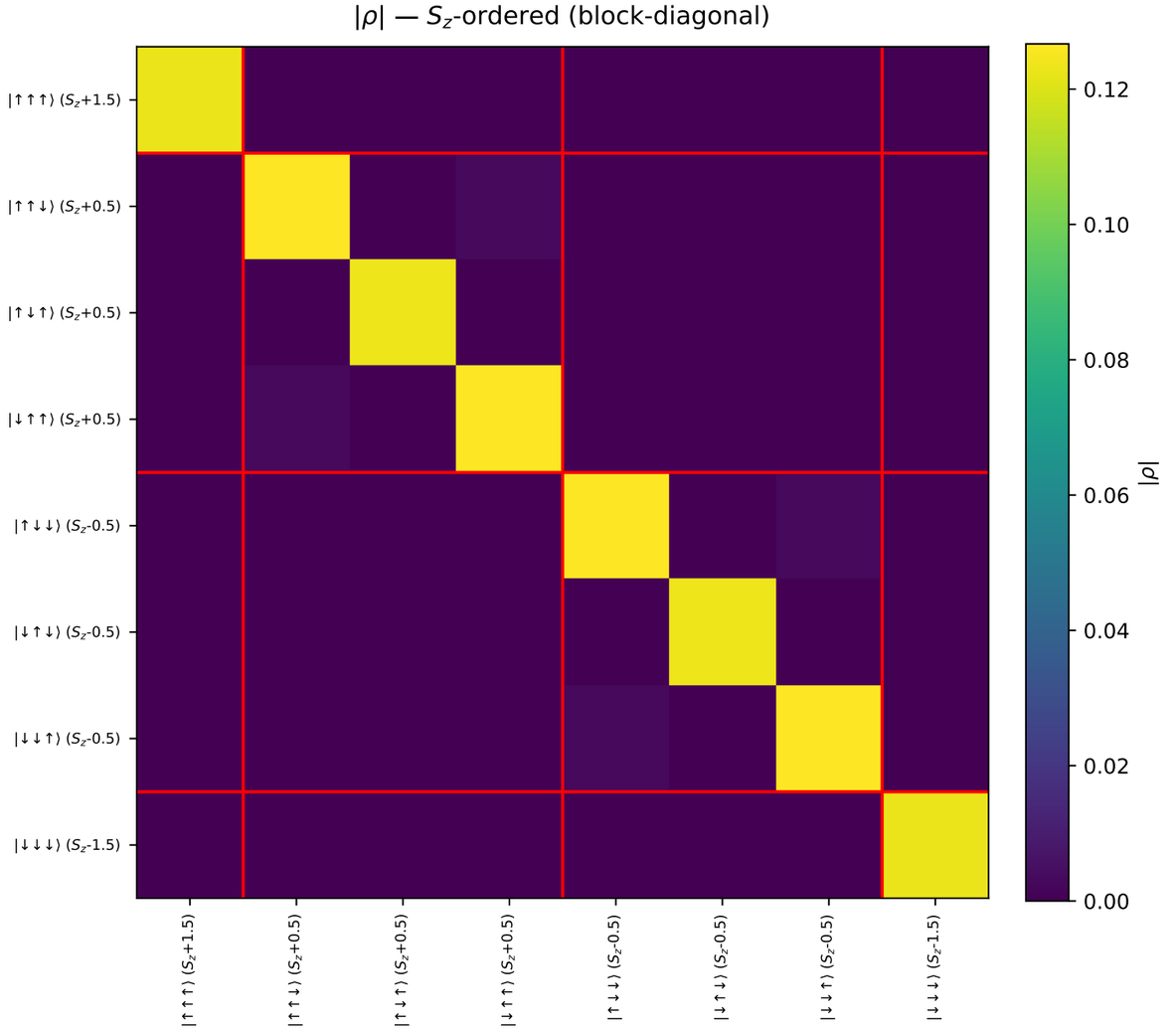
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.110237  C_GME=0.000000
2. p=0.123020  C_GME=0.000000
3. p=0.112657  C_GME=0.000099
4. p=0.100769  C_GME=0.000000
5. p=0.110541  C_GME=0.000000
... and 5 more states
```

3-Body RDM (Sites 0,1,5)





Trace=1.000000 Purity=0.125067 Entropy=2.079175 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,1,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1250673282
  Von Neumann entropy  : 2.0791745511
  Eigenvalue min       : 1.231782e-01
  Eigenvalue max       : 1.300146e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12506733

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

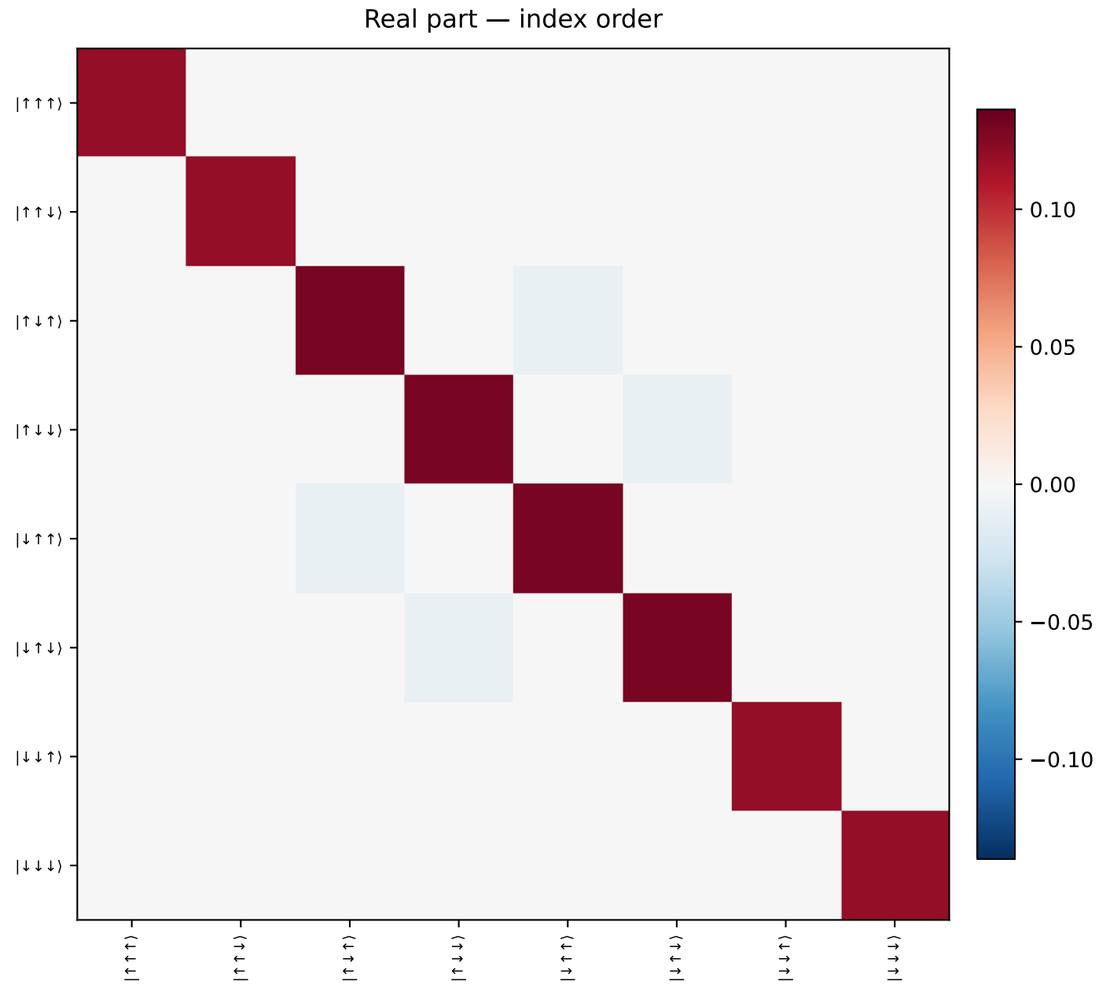
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.41704627

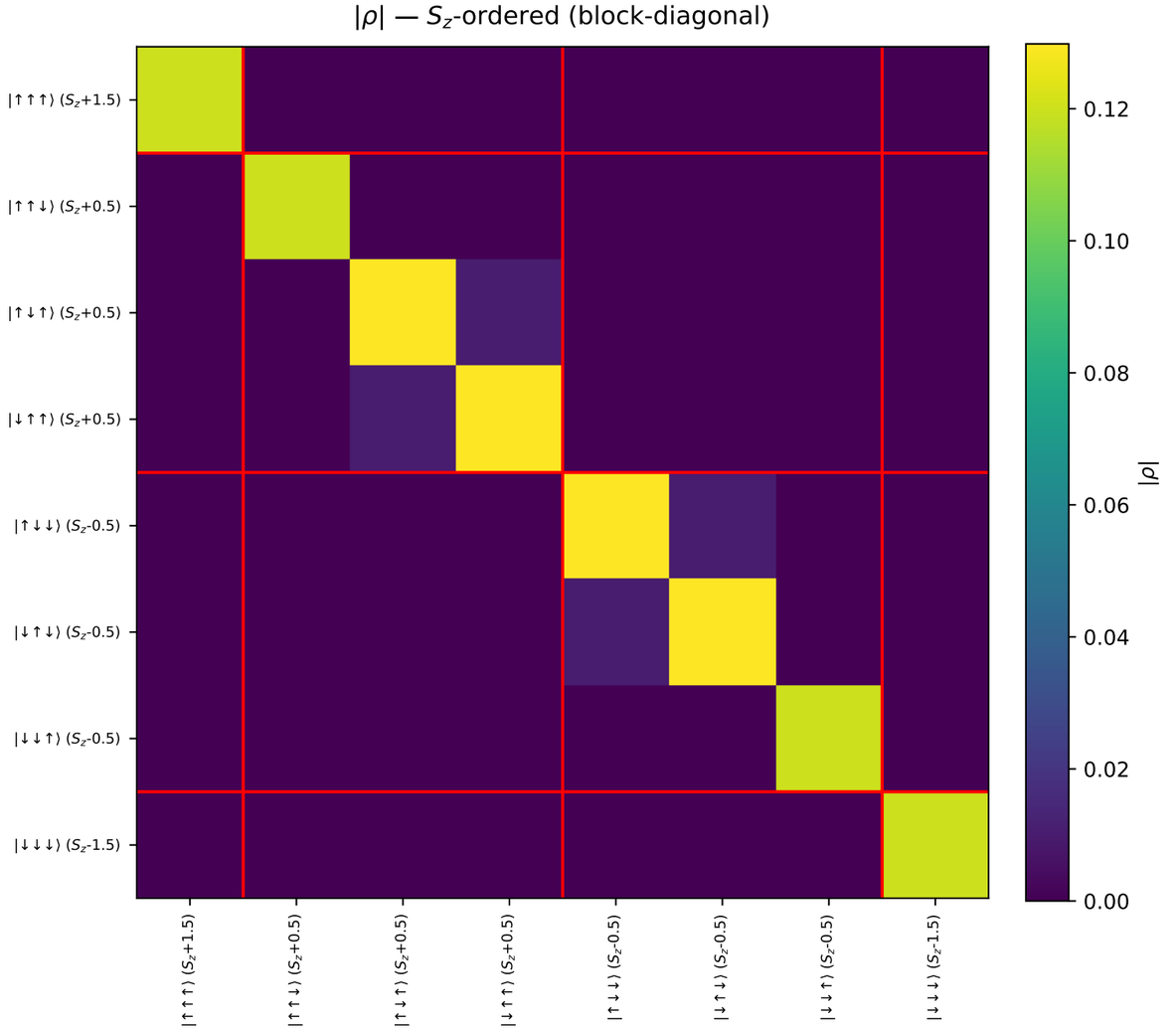
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.061032  C_GME=0.000076
2. p=0.095216  C_GME=0.000787
3. p=0.093016  C_GME=0.000000
4. p=0.093941  C_GME=0.000000
5. p=0.113110  C_GME=0.001561
... and 5 more states
```

3-Body RDM (Sites 0,2,3)





Trace=1.000000 Purity=0.125559 Entropy=2.077261 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1255585209
  Von Neumann entropy  : 2.0772612935
  Eigenvalue min       : 1.201759e-01
  Eigenvalue max       : 1.394722e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12555852

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

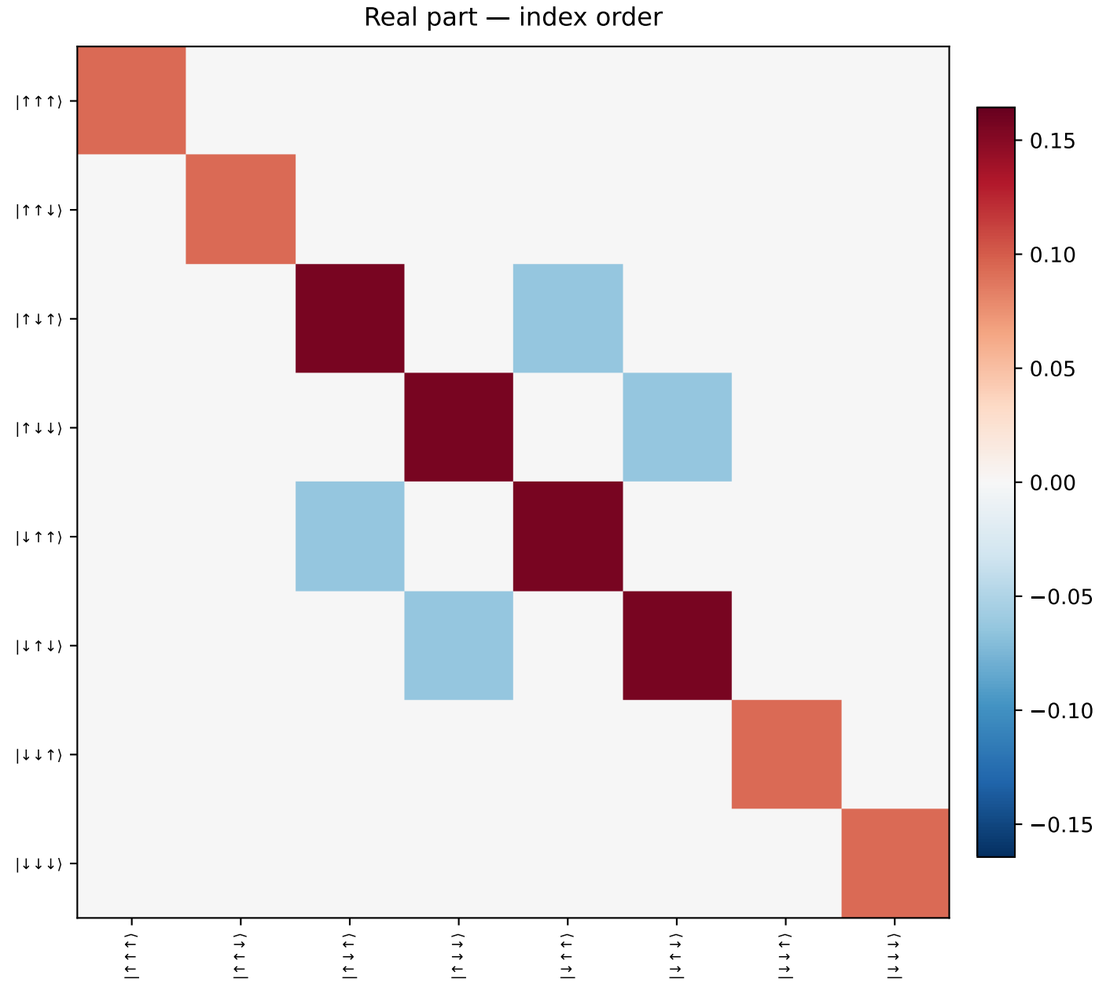
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000001

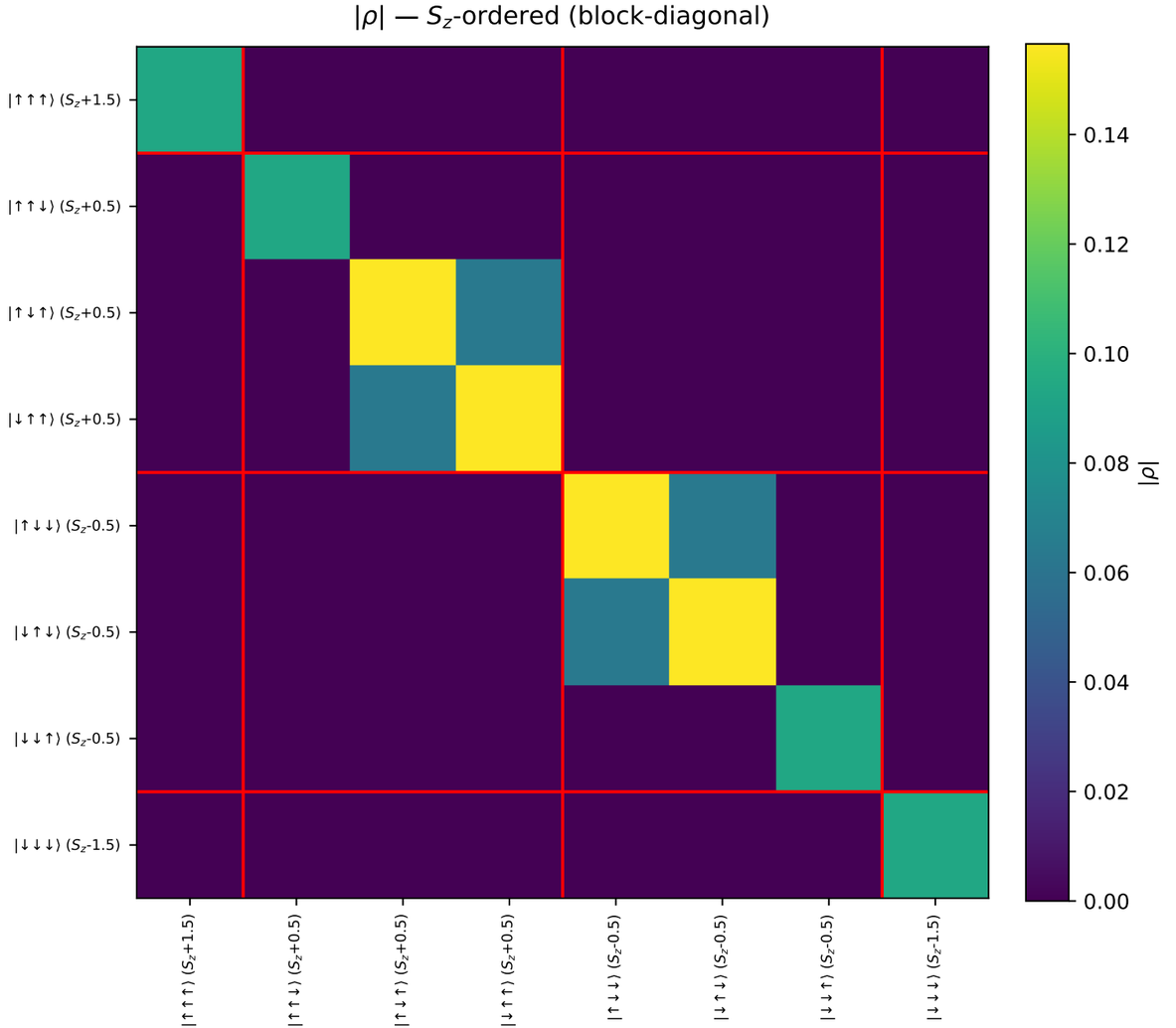
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.104433  C_GME=0.000000
2. p=0.085273  C_GME=0.000006
3. p=0.100016  C_GME=0.000000
4. p=0.119682  C_GME=0.000000
5. p=0.109507  C_GME=0.000000
... and 5 more states
```

3-Body RDM (Sites 0,2,4)





Trace=1.000000 Purity=0.148904 Entropy=1.994851 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,2,4)

RDM Metrics

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.1489040060
Von Neumann entropy	1.9948505094
Eigenvalue min	9.344053e-02
Eigenvalue max	2.196784e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis

gamma = Tr(rho^2) = 0.14890401

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis

Lower Bounds:
- C_LB = 0.00000000

C_GME (unitary optimization):
- C_stiefel (unitary opt) = nan BEST

C_GME (unitary optimization - Expoenetial Map):

Upper Bound:
- C_UB (eigendecomp) = 0.00000001

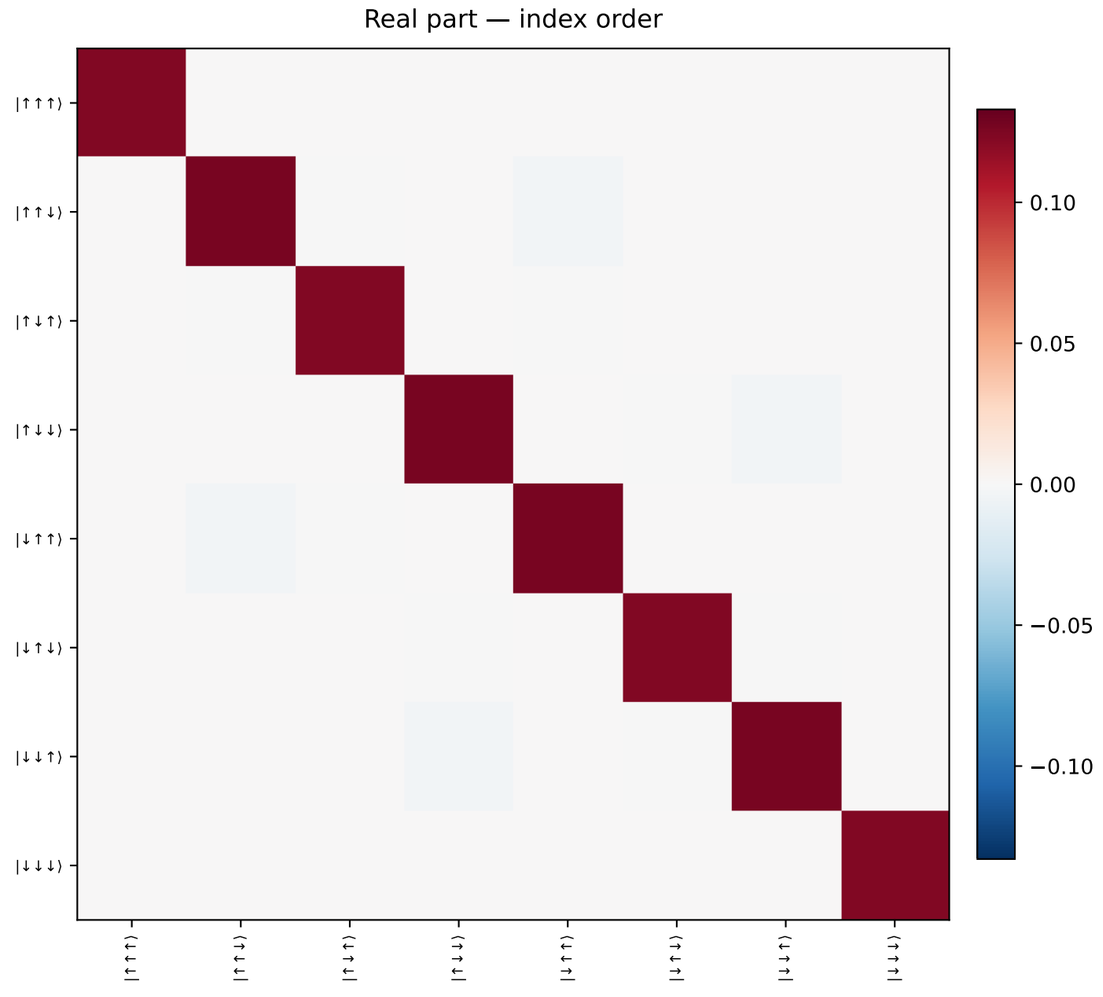
Status: inconclusive

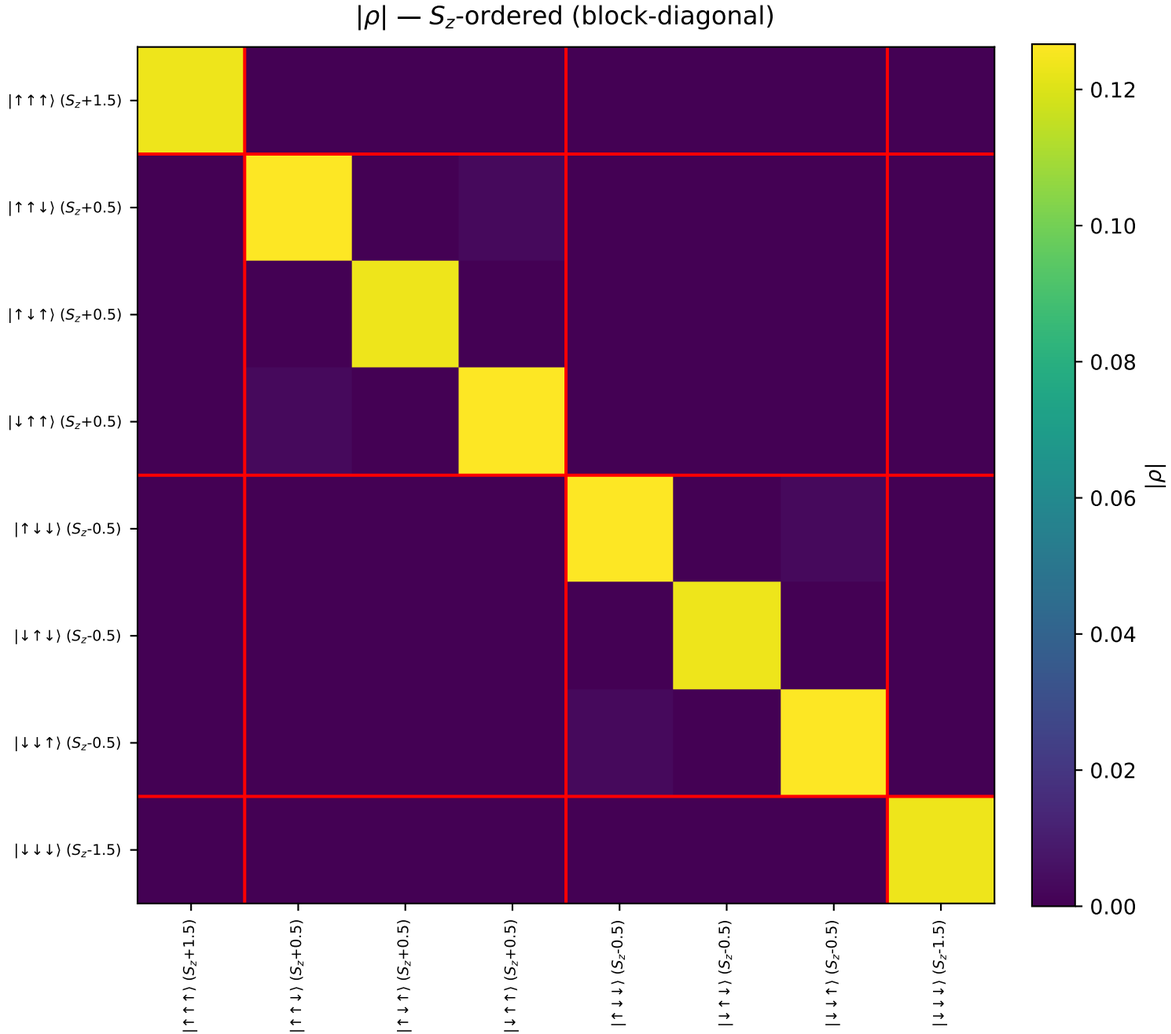
Solver source:
- –

1. p=0.097317 C_GME=0.000000
2. p=0.108378 C_GME=0.000000
3. p=0.081953 C_GME=0.000000
4. p=0.104202 C_GME=0.000000
5. p=0.074520 C_GME=0.001232

... and 5 more states

3-Body RDM (Sites 0,2,5)





Trace=1.000000 Purity=0.125065 Entropy=2.079184 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,2,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1250649358
  Von Neumann entropy  : 2.0791839988
  Eigenvalue min       : 1.232101e-01
  Eigenvalue max       : 1.299246e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12506494

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

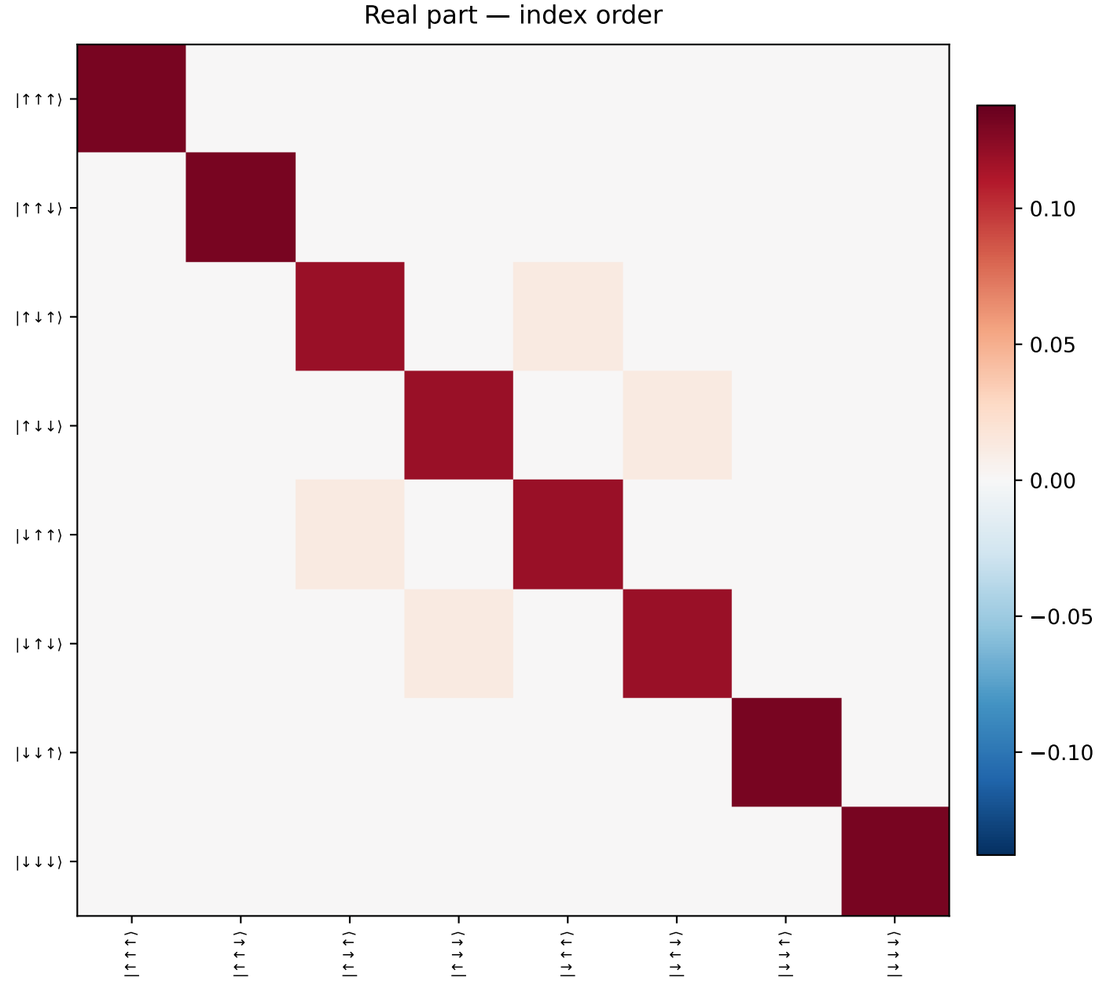
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.41707374

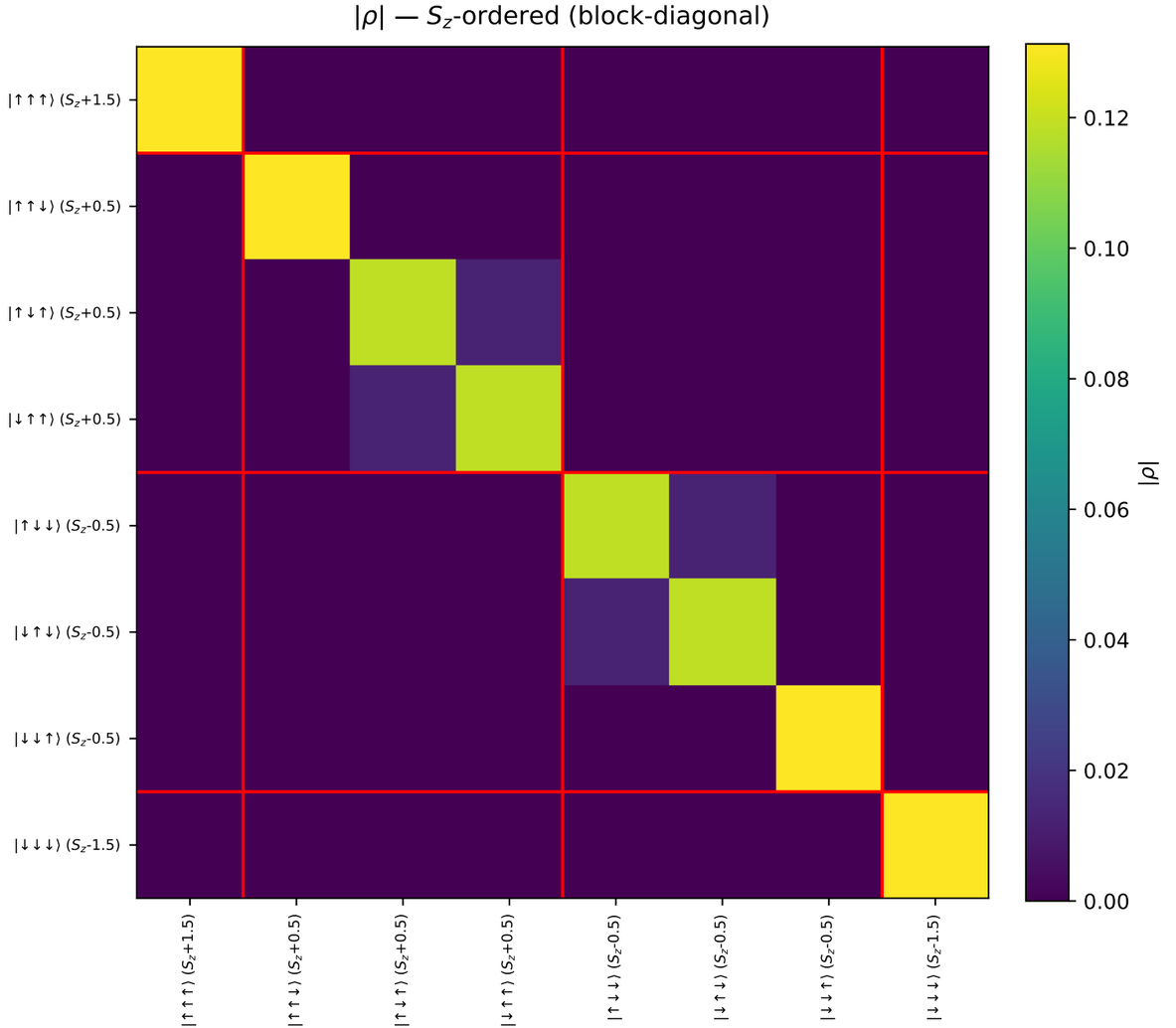
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.118736  C_GME=0.004047
2. p=0.104181  C_GME=0.000000
3. p=0.099722  C_GME=0.000000
4. p=0.098833  C_GME=0.000000
5. p=0.083866  C_GME=0.000000
... and 5 more states
```

3-Body RDM (Sites 0,3,4)





Trace=1.000000 Purity=0.125949 Entropy=2.075505 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1259492576
  Von Neumann entropy  : 2.0755048607
  Eigenvalue min       : 1.061328e-01
  Eigenvalue max       : 1.312891e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12594926

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

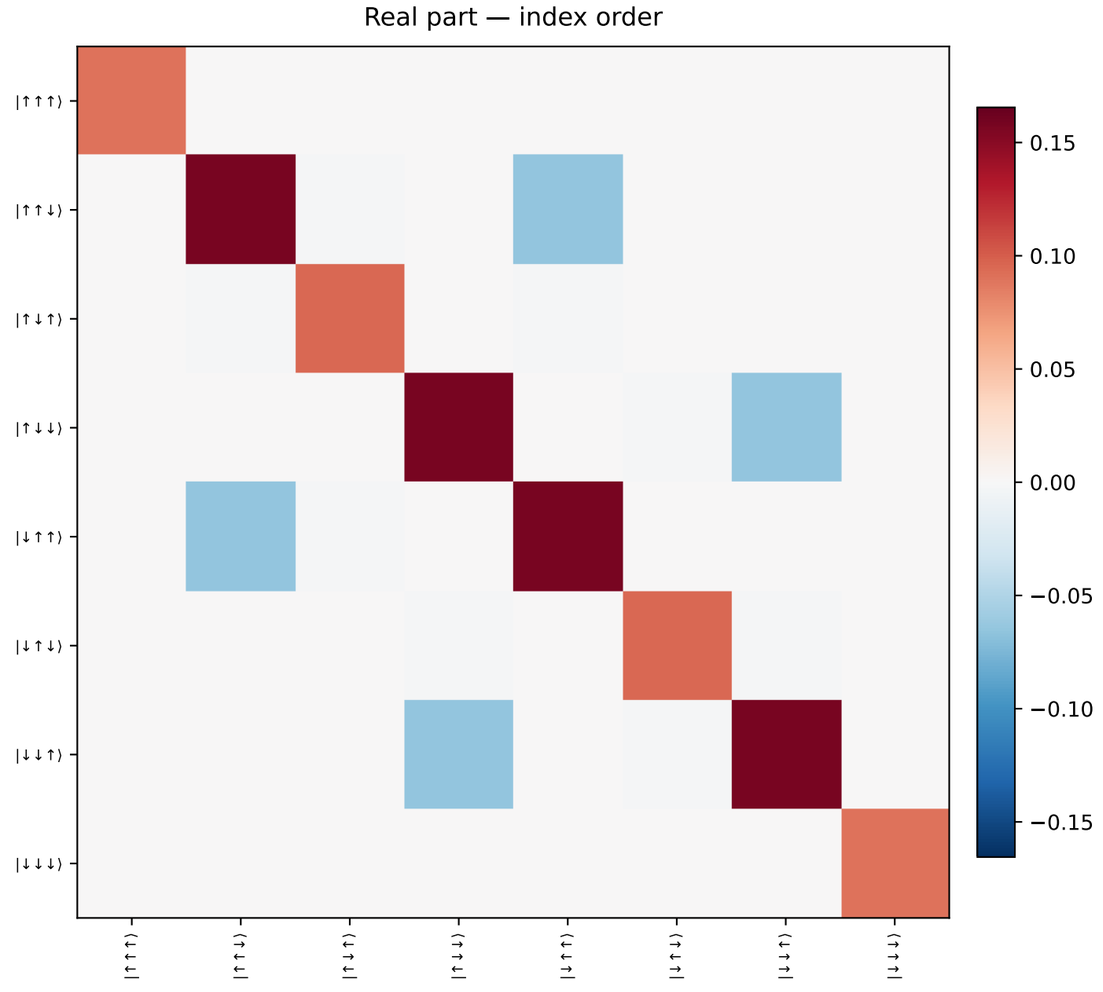
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.00000001

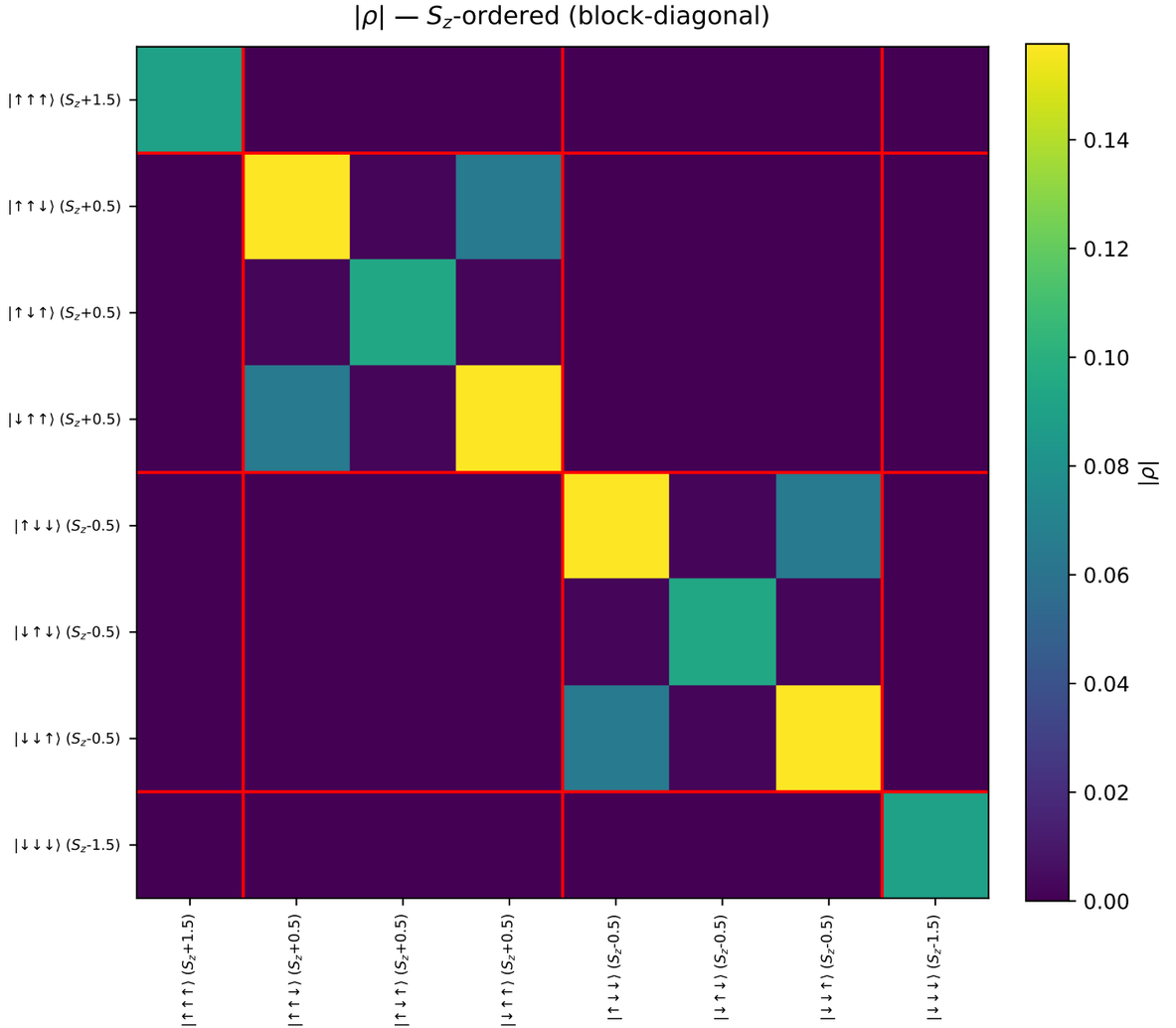
  Status: no_GME

  Solver source:
    –
```

```
1. p=0.078294  C_GME=0.000000
2. p=0.109383  C_GME=0.000000
3. p=0.099869  C_GME=0.000000
4. p=0.096236  C_GME=0.000000
5. p=0.097404  C_GME=0.000000
... and 5 more states
```

3-Body RDM (Sites 0,3,5)





Trace=1.000000 Purity=0.150430 Entropy=1.989597 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,3,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1504300892
  Von Neumann entropy  : 1.9895972634
  Eigenvalue min       : 9.049354e-02
  Eigenvalue max       : 2.225630e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.15043009

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

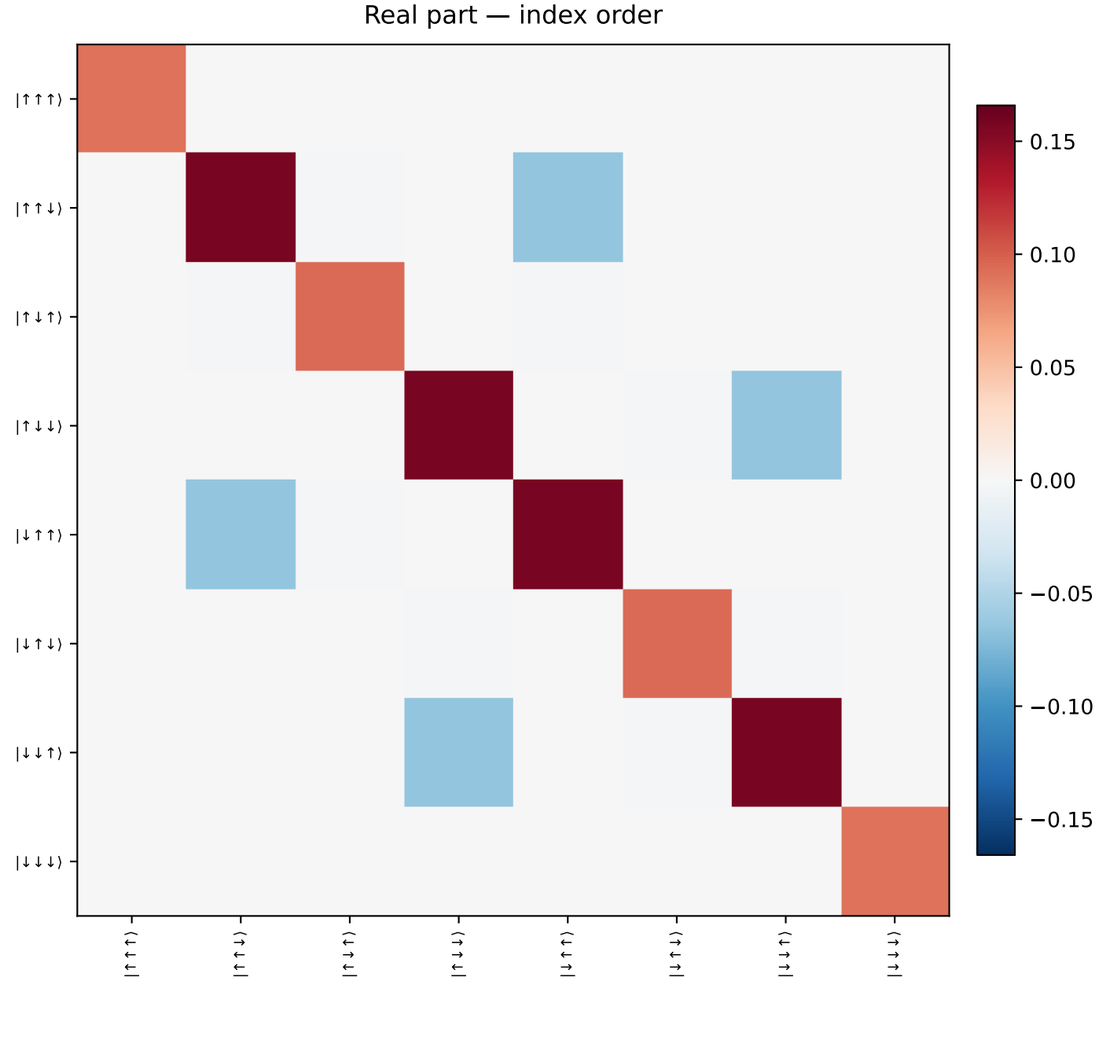
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.31520009

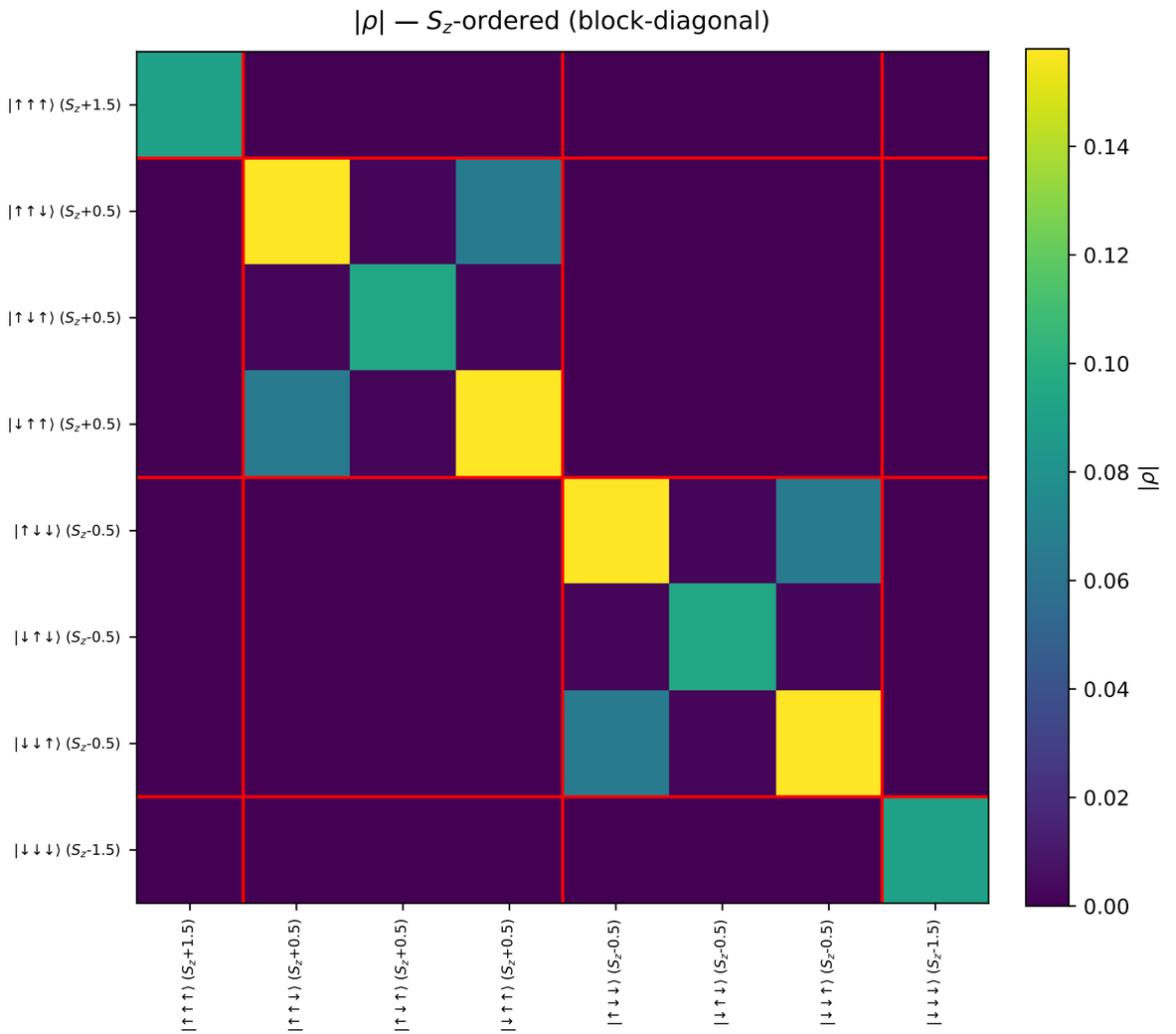
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.079091  C_GME=0.000000
2. p=0.097178  C_GME=0.000000
3. p=0.083970  C_GME=0.000000
4. p=0.074451  C_GME=0.000003
5. p=0.143075  C_GME=0.000000
... and 5 more states
```

3-Body RDM (Sites 0,4,5)





Trace=1.000000 Purity=0.150975 Entropy=1.987757 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,4,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1509750785
  Von Neumann entropy  : 1.9877574325
  Eigenvalue min       : 9.014436e-02
  Eigenvalue max       : 2.236047e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.15097508

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

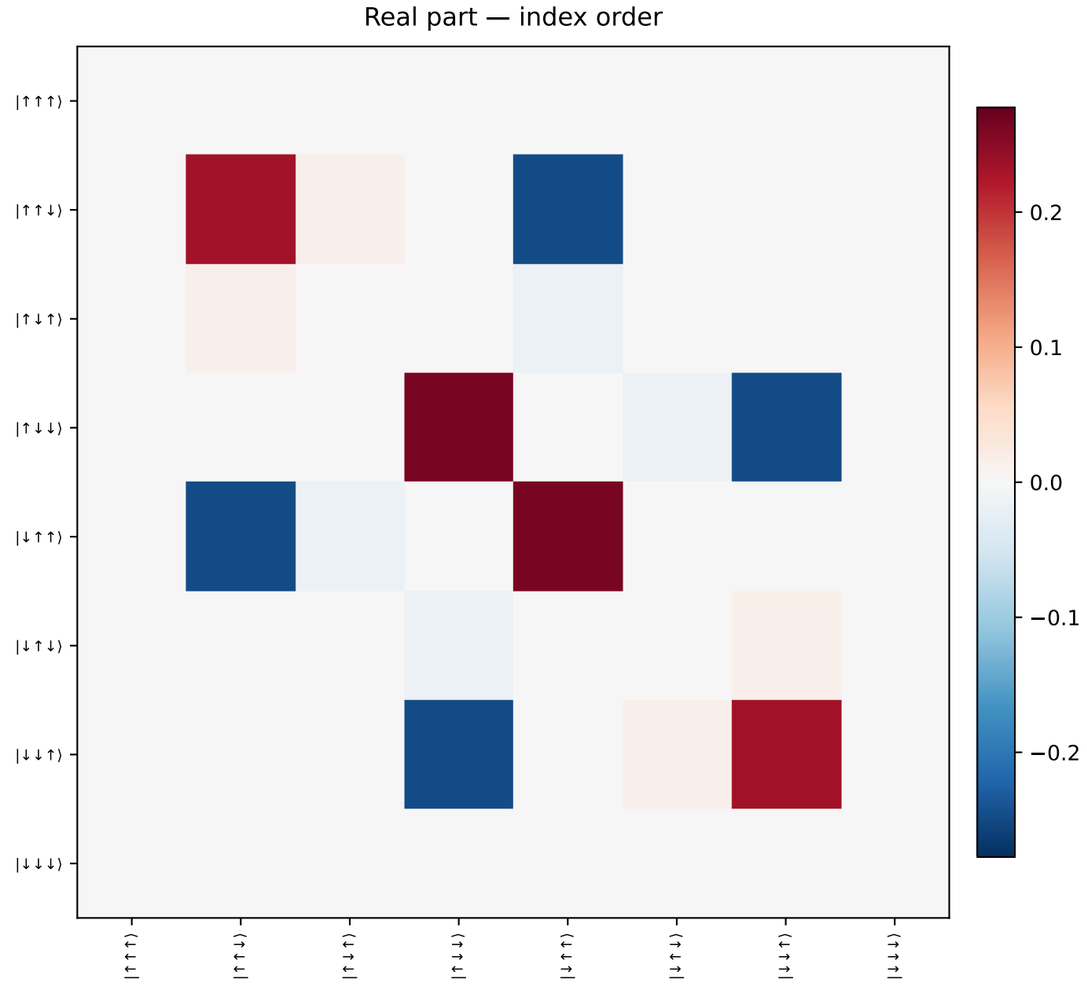
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.31402822

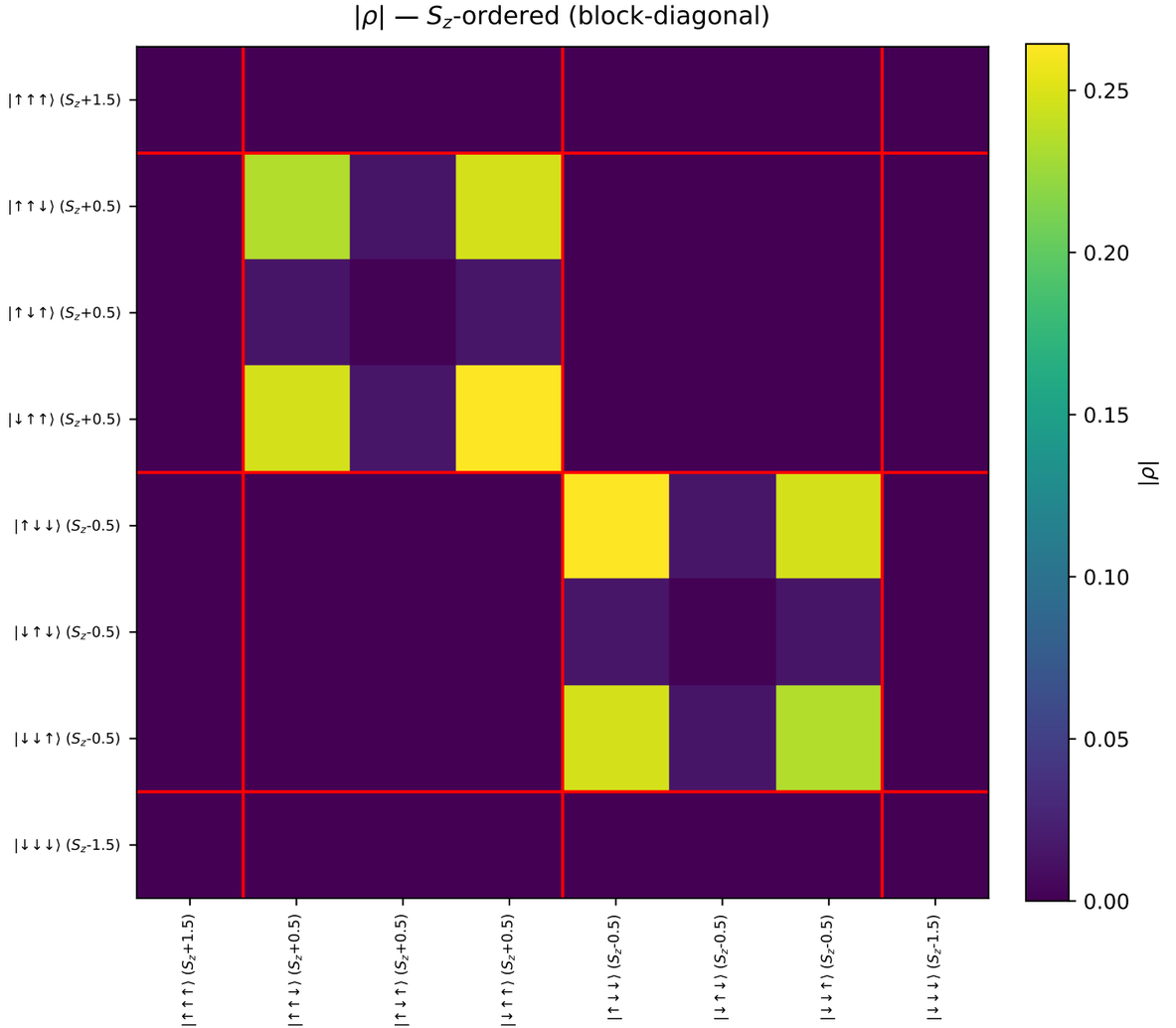
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.098243  C_GME=0.000006
2. p=0.076169  C_GME=0.004062
3. p=0.124309  C_GME=0.000004
4. p=0.113437  C_GME=0.000000
5. p=0.035212  C_GME=0.000000
... and 5 more states
```

3-Body RDM (Sites 1,2,3)





Trace=1.000000 Purity=0.496207 Entropy=0.722312 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,2,3)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4962065490
  Von Neumann entropy  : 0.7223117676
  Eigenvalue min       : 6.338489e-04
  Eigenvalue max       : 4.980985e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.49620655

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

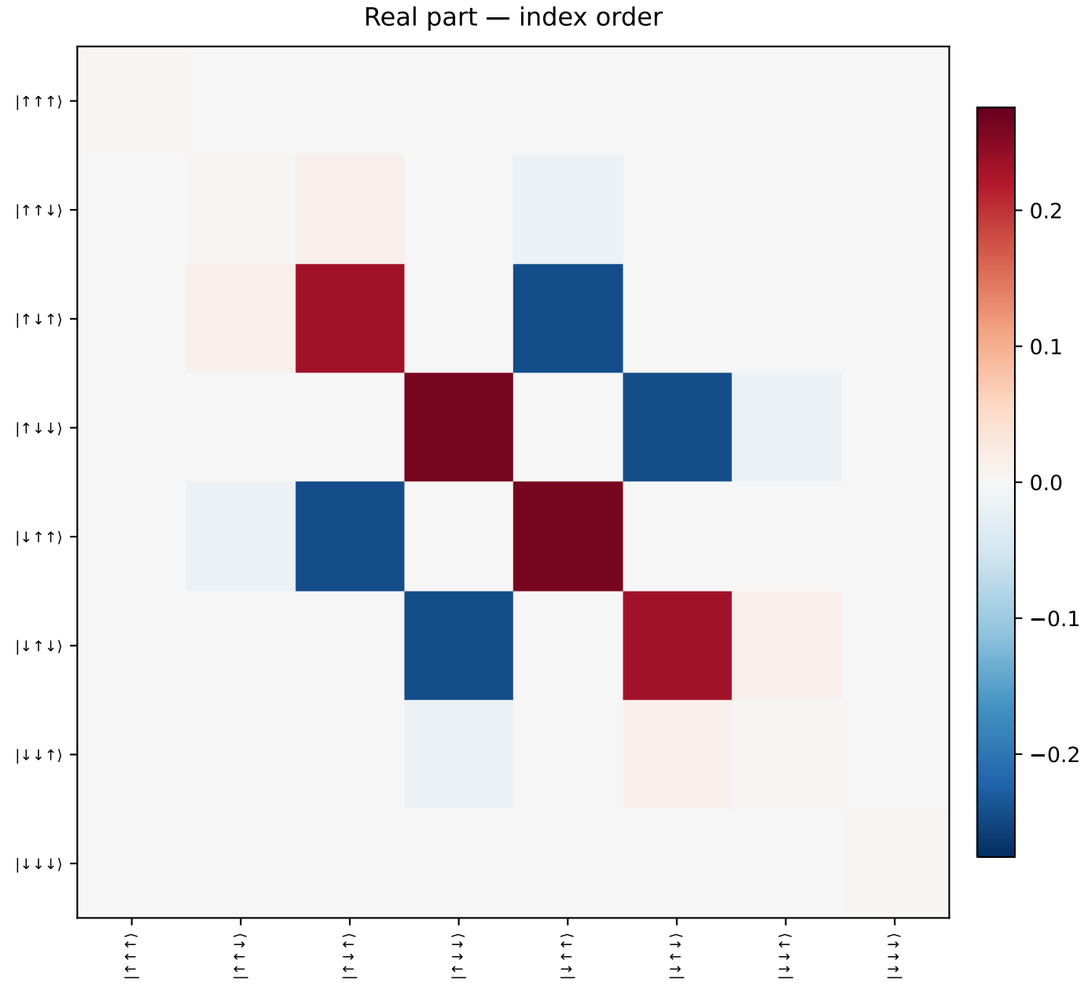
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.08896268

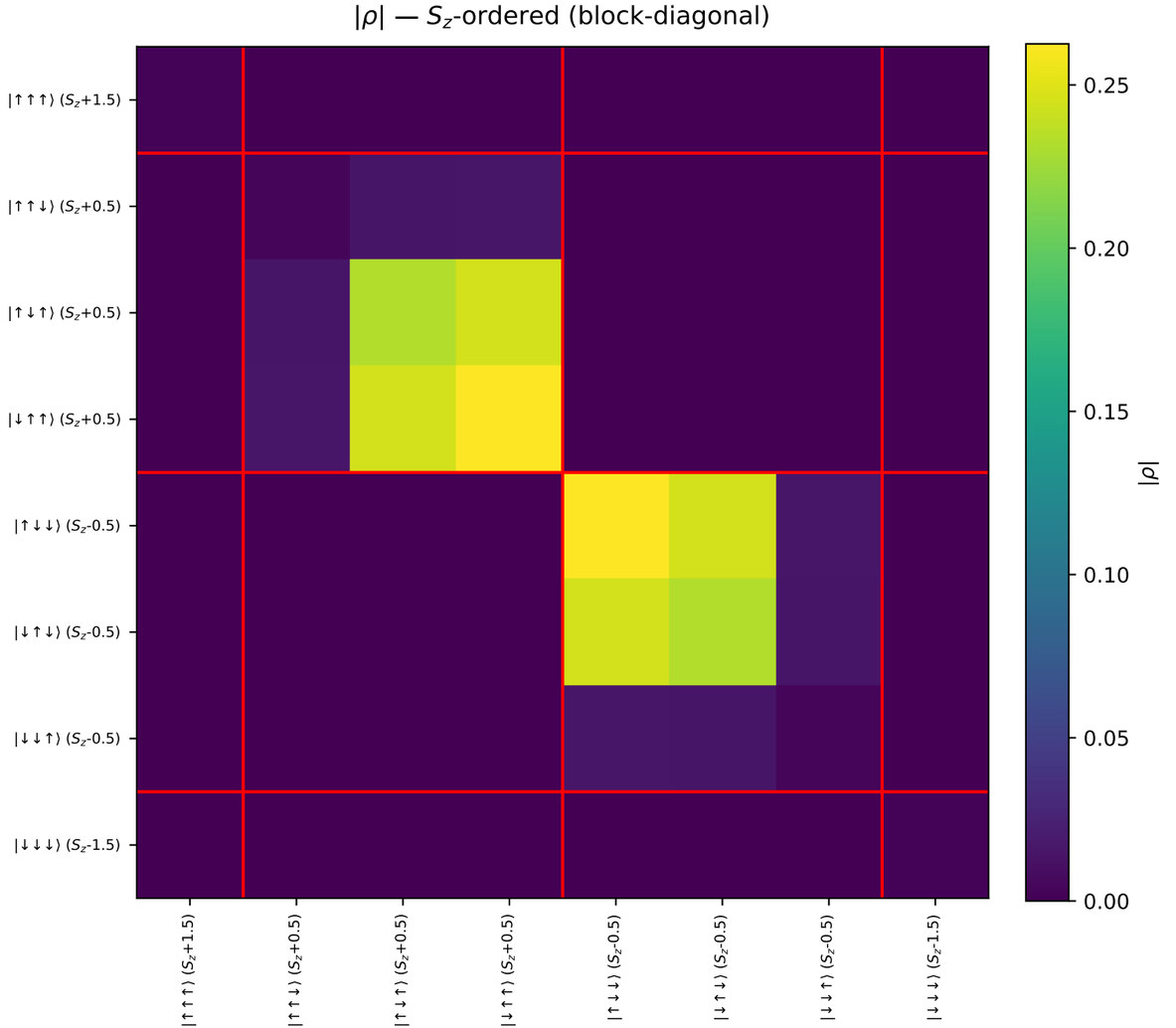
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.014067  C_GME=0.358505
2. p=0.157274  C_GME=0.027355
3. p=0.162682  C_GME=0.031762
4. p=0.023549  C_GME=0.019195
5. p=0.135504  C_GME=0.023572
... and 5 more states
```

3-Body RDM (Sites 1,2,4)





Trace=1.000000 Purity=0.486887 Entropy=0.778046 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,2,4)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4868869169
  Von Neumann entropy  : 0.7780459582
  Eigenvalue min       : 2.204961e-03
  Eigenvalue max       : 4.933851e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.48688692

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

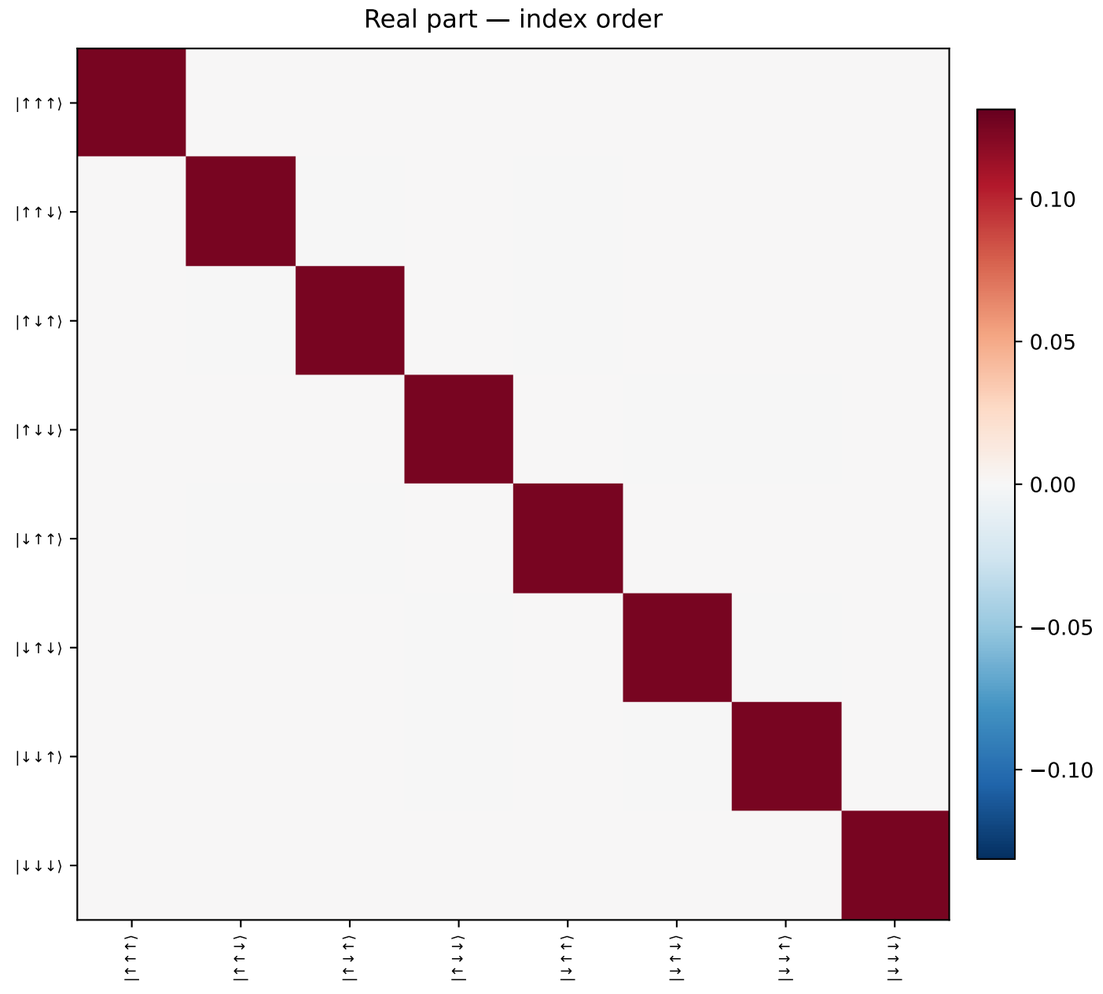
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.08851678

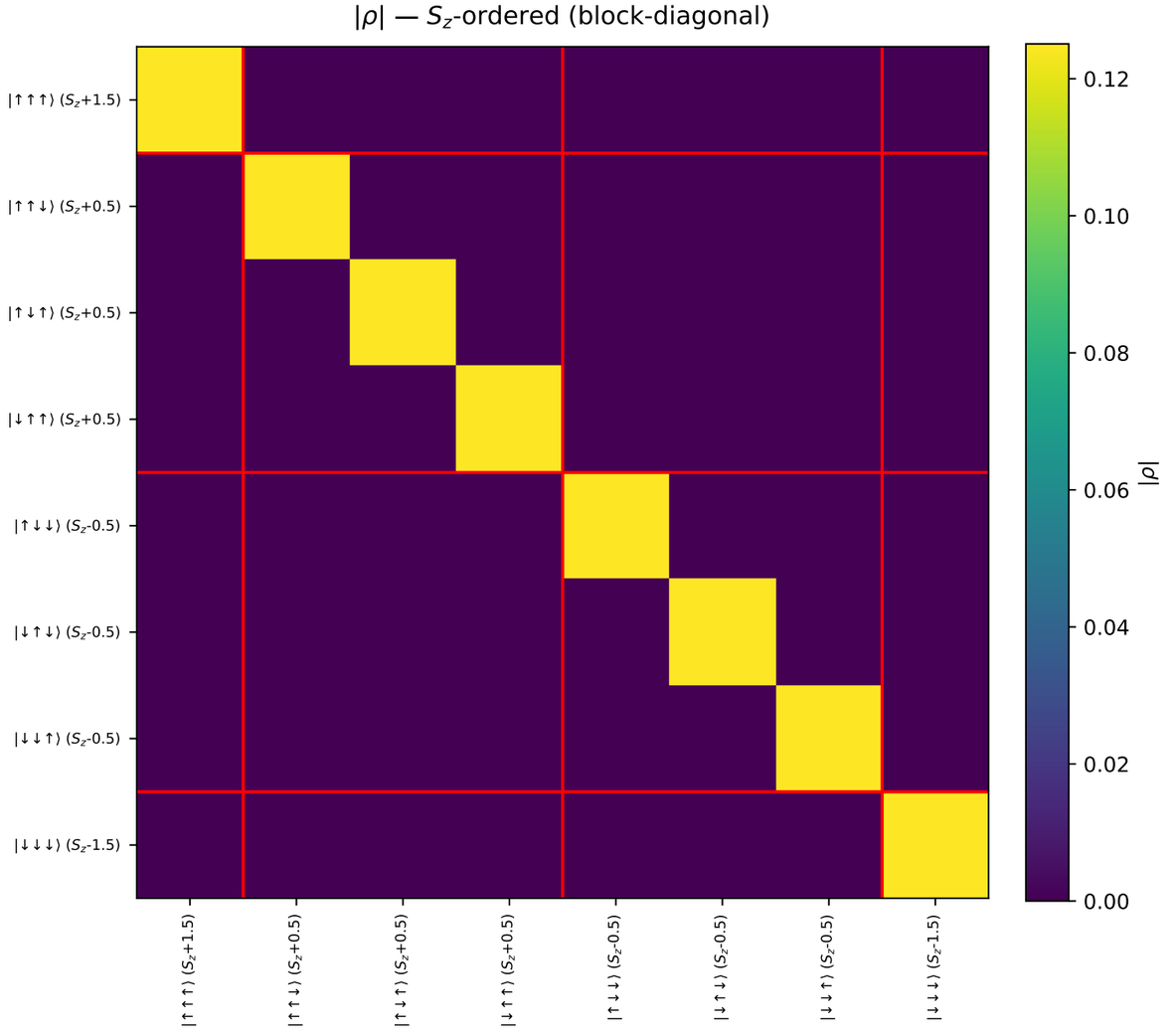
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.162555  C_GME=0.000000
2. p=0.131407  C_GME=0.000000
3. p=0.150347  C_GME=0.000000
4. p=0.117094  C_GME=0.000017
5. p=0.221171  C_GME=0.000000
... and 5 more states
```

3-Body RDM (Sites 1,2,5)





Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,2,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1250001535
  Von Neumann entropy  : 2.0794409280
  Eigenvalue min       : 1.249047e-01
  Eigenvalue max       : 1.252374e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12500015

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

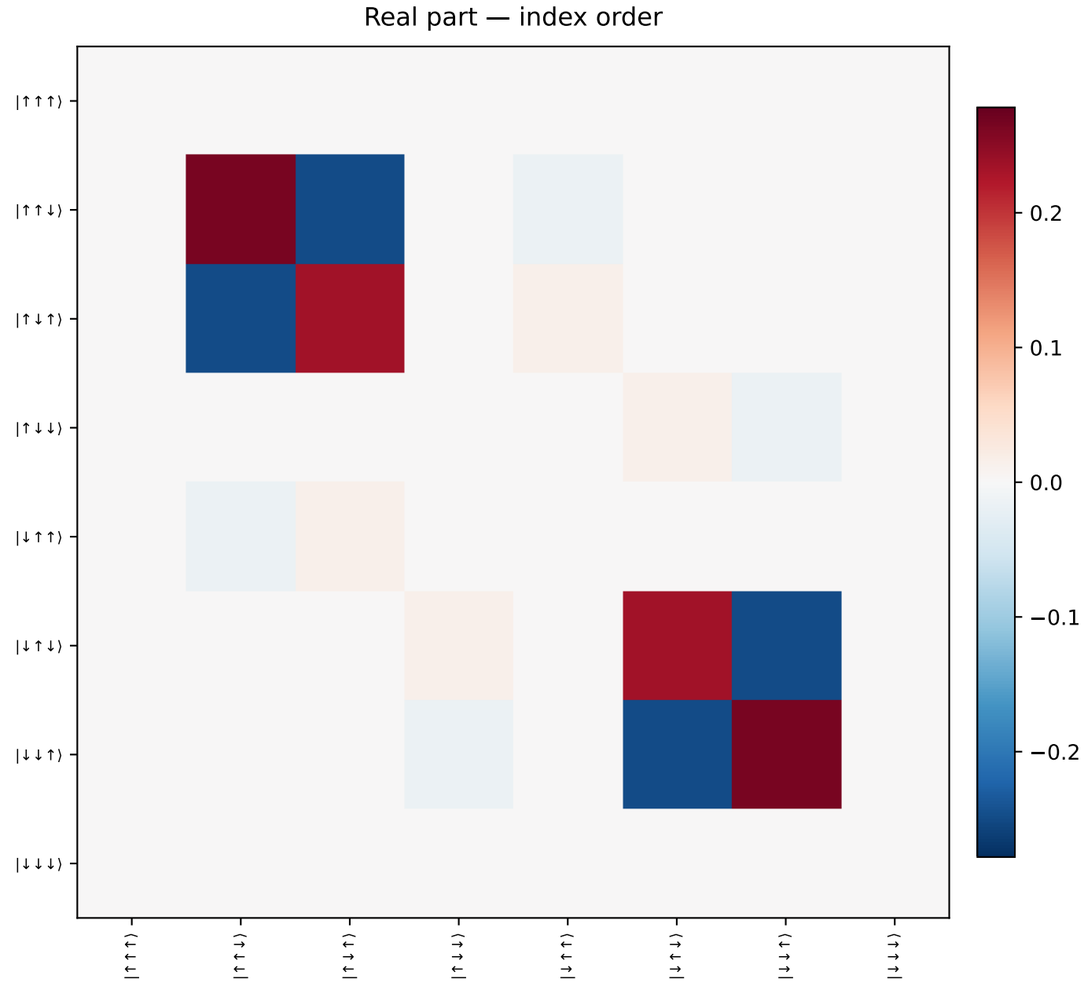
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.42282845

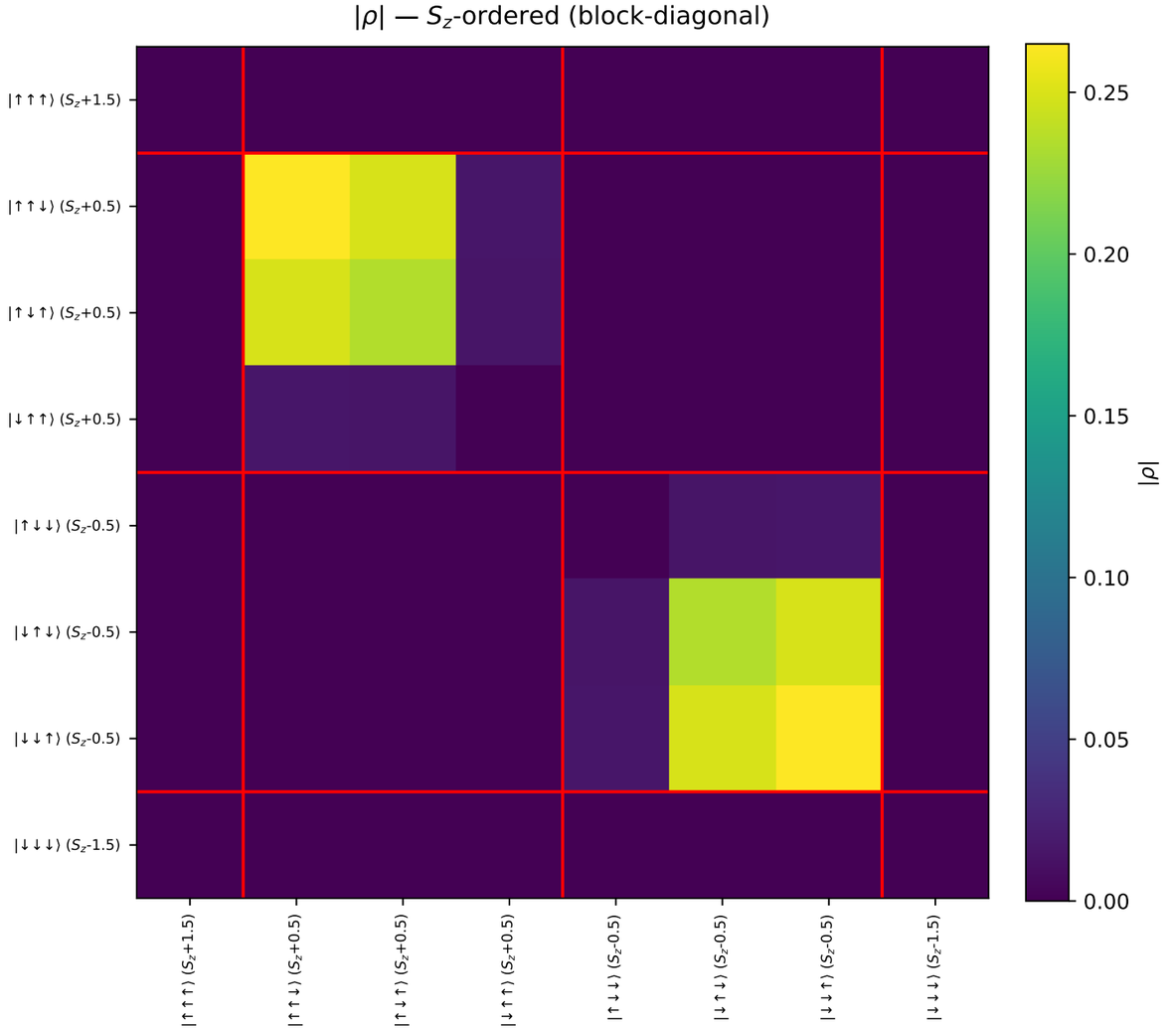
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.071684  C_GME=0.000000
2. p=0.070691  C_GME=0.000000
3. p=0.116500  C_GME=0.000001
4. p=0.113211  C_GME=0.017731
5. p=0.105202  C_GME=0.000008
... and 5 more states
```

3-Body RDM (Sites 1,3,4)





Trace=1.000000 Purity=0.499664 Entropy=0.696539 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,3,4)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4996641682
  Von Neumann entropy  : 0.6965387921
  Eigenvalue min       : 5.510387e-05
  Eigenvalue max       : 4.998320e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.49966417

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
==================================

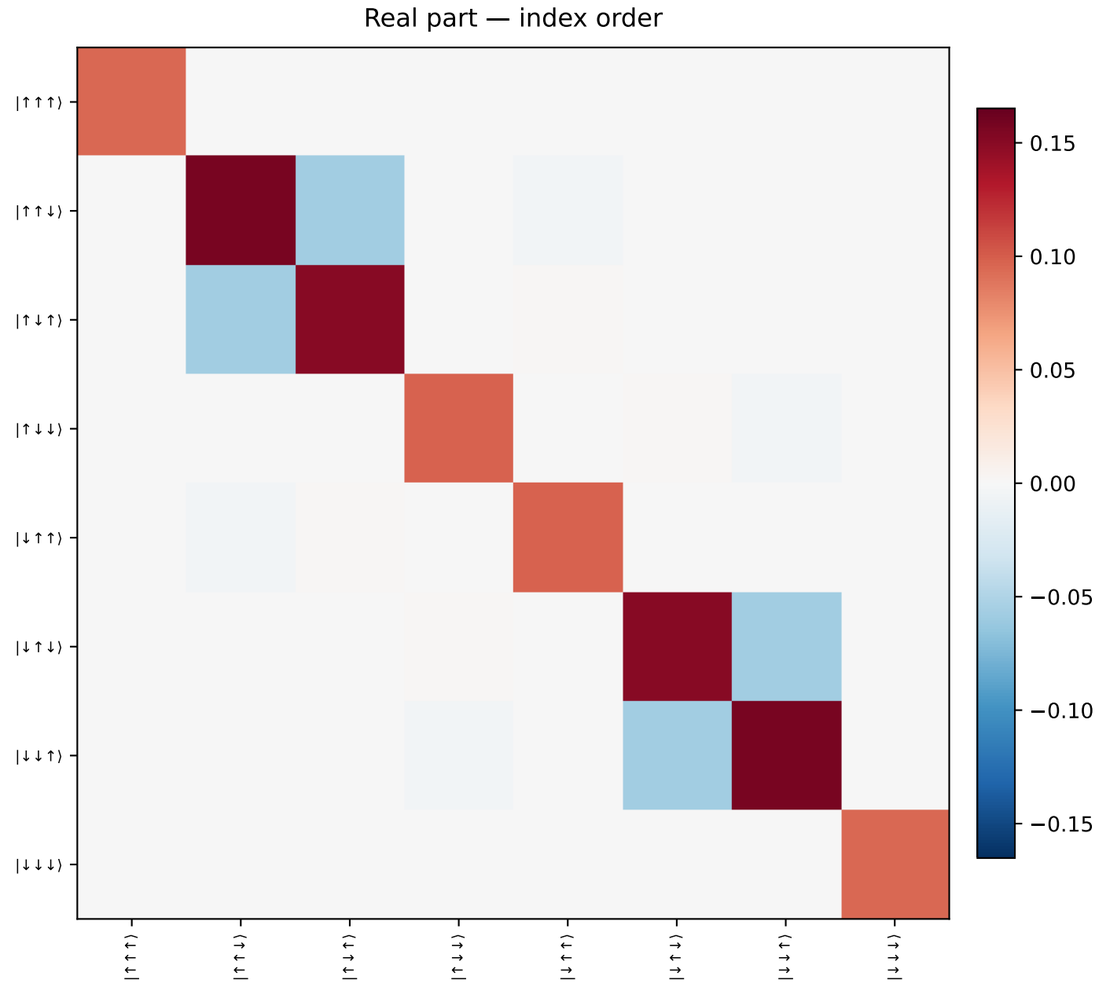
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.08794920

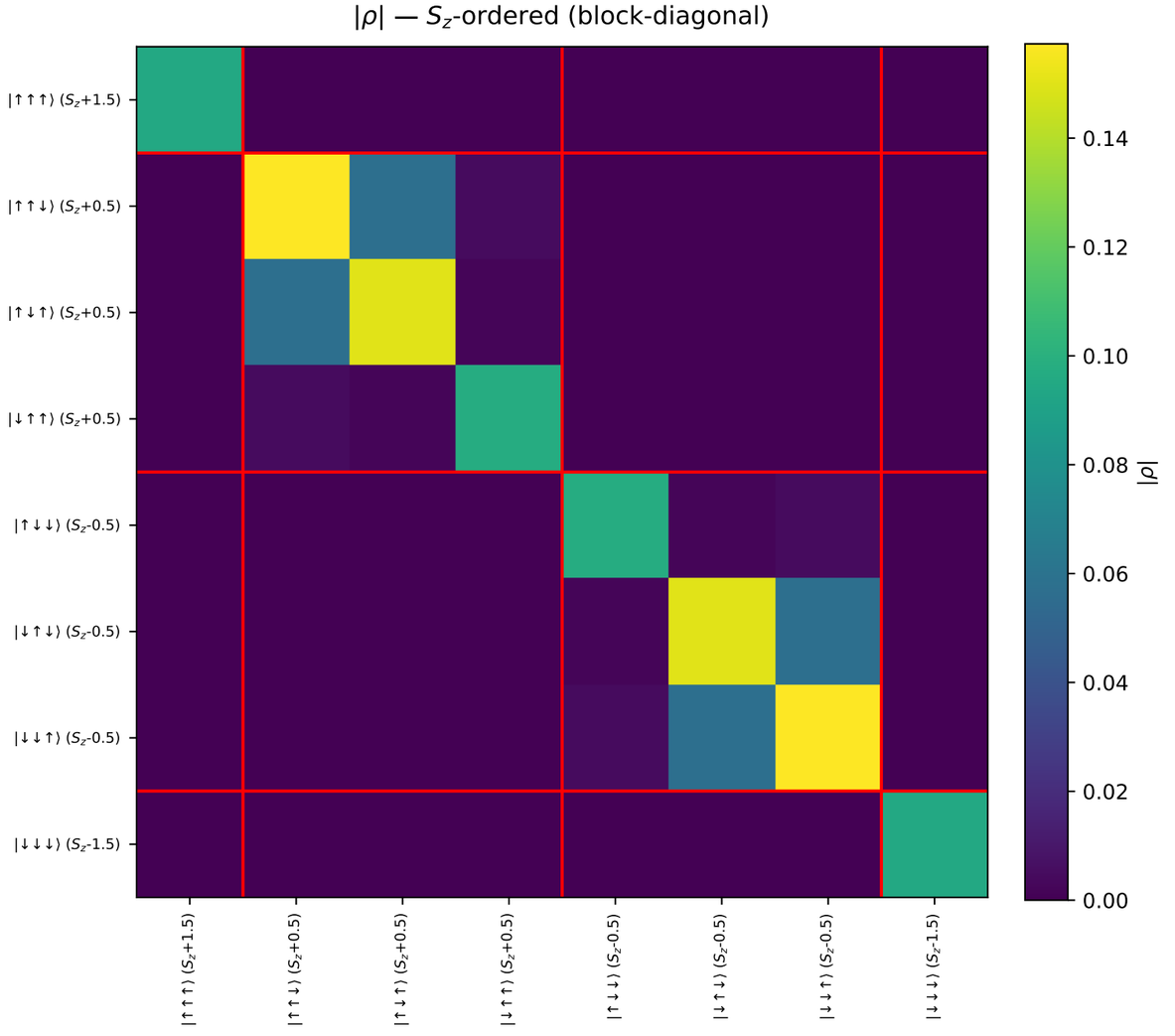
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.165830  C_GME=0.068826
2. p=0.028059  C_GME=0.022001
3. p=0.137861  C_GME=0.083703
4. p=0.120177  C_GME=0.052682
5. p=0.119108  C_GME=0.070427
... and 4 more states
```

3-Body RDM (Sites 1,3,5)





Trace=1.000000 Purity=0.145172 Entropy=2.007503 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,3,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1451716469
  Von Neumann entropy  : 2.0075033507
  Eigenvalue min       : 9.470402e-02
  Eigenvalue max       : 2.119283e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.14517165

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

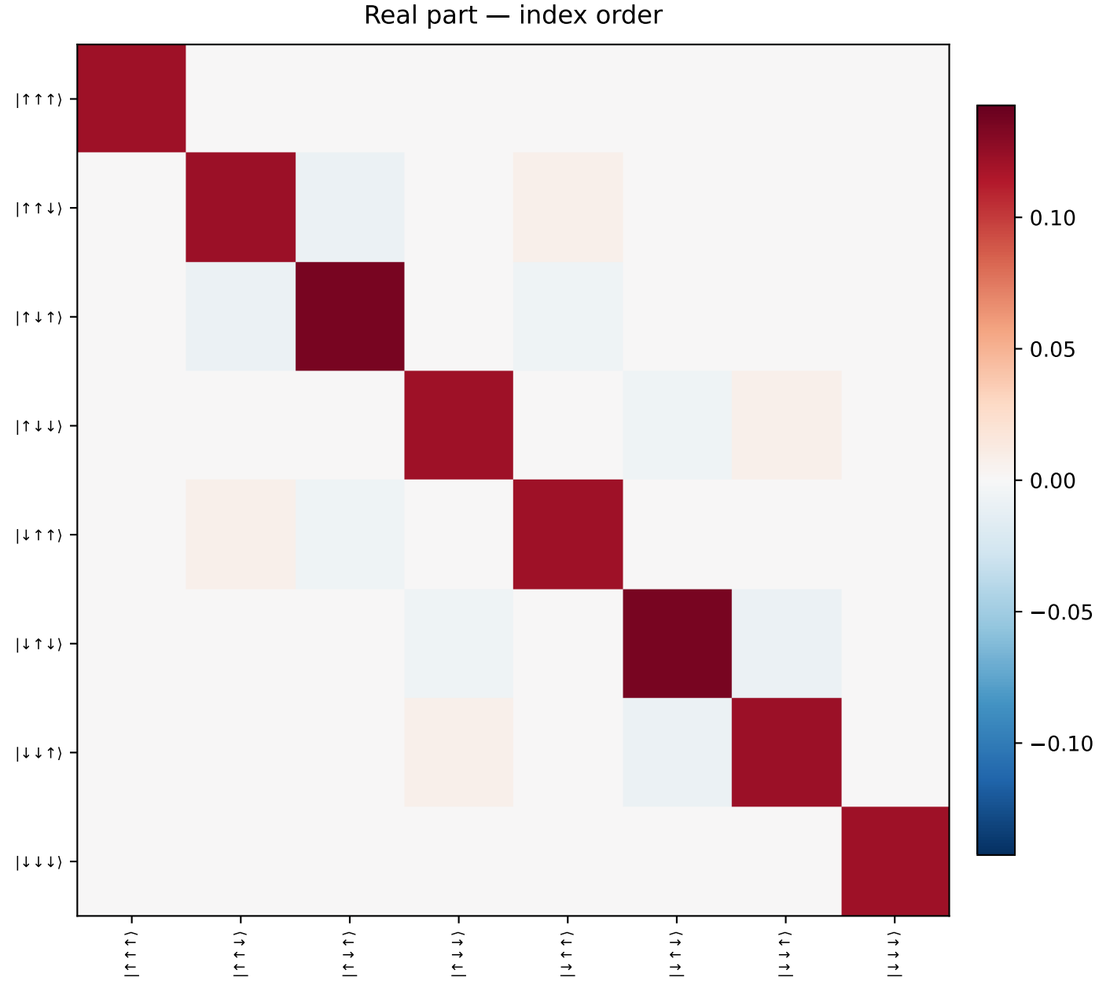
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.35078290

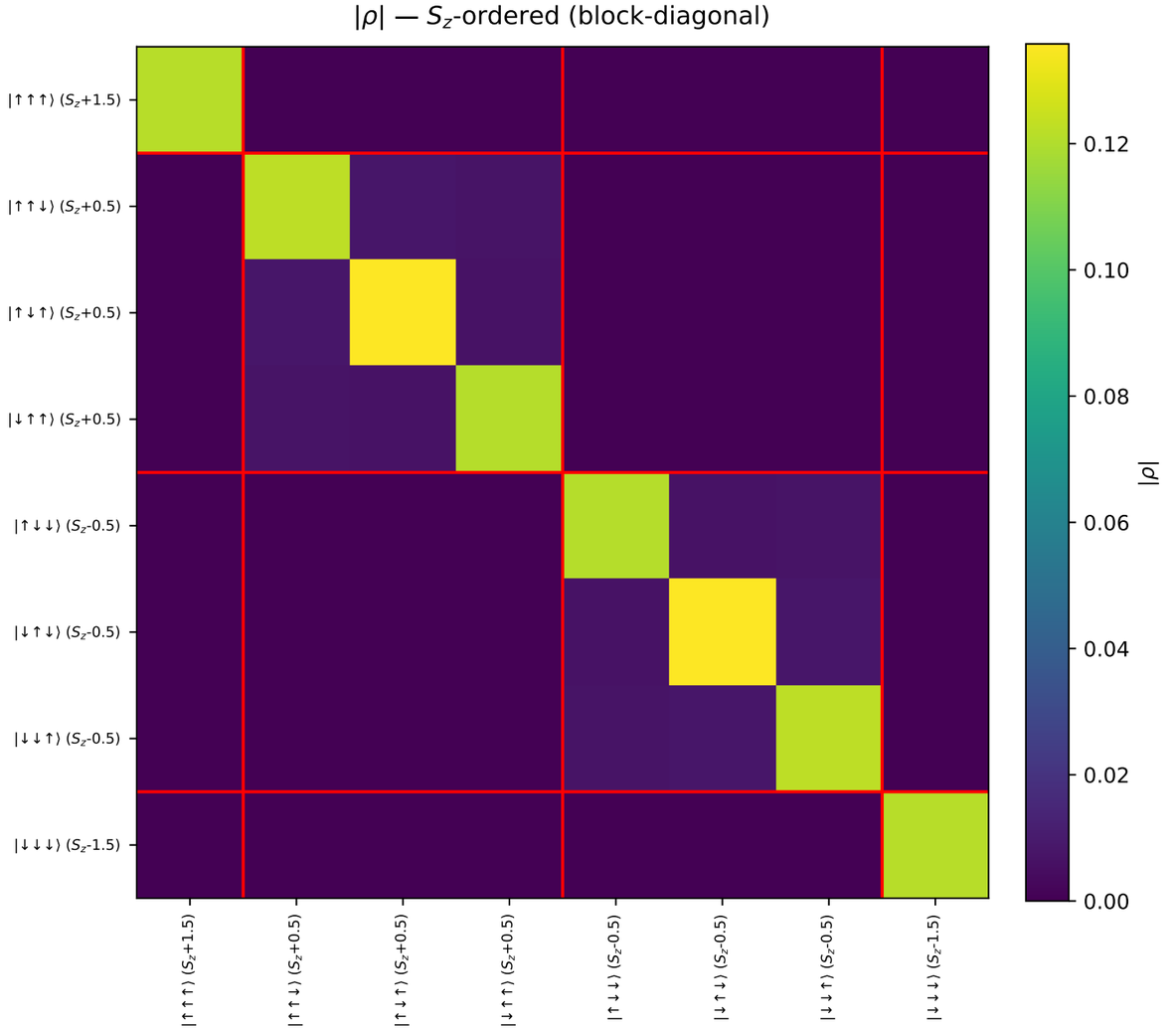
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.103500  C_GME=0.000009
2. p=0.118069  C_GME=0.000001
3. p=0.134178  C_GME=0.000091
4. p=0.077887  C_GME=0.000004
5. p=0.100848  C_GME=0.000001
... and 5 more states
```

3-Body RDM (Sites 1,4,5)





Trace=1.000000 Purity=0.125941 Entropy=2.075762 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1259414218
  Von Neumann entropy  : 2.0757624647
  Eigenvalue min       : 1.141110e-01
  Eigenvalue max       : 1.430661e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12594142

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

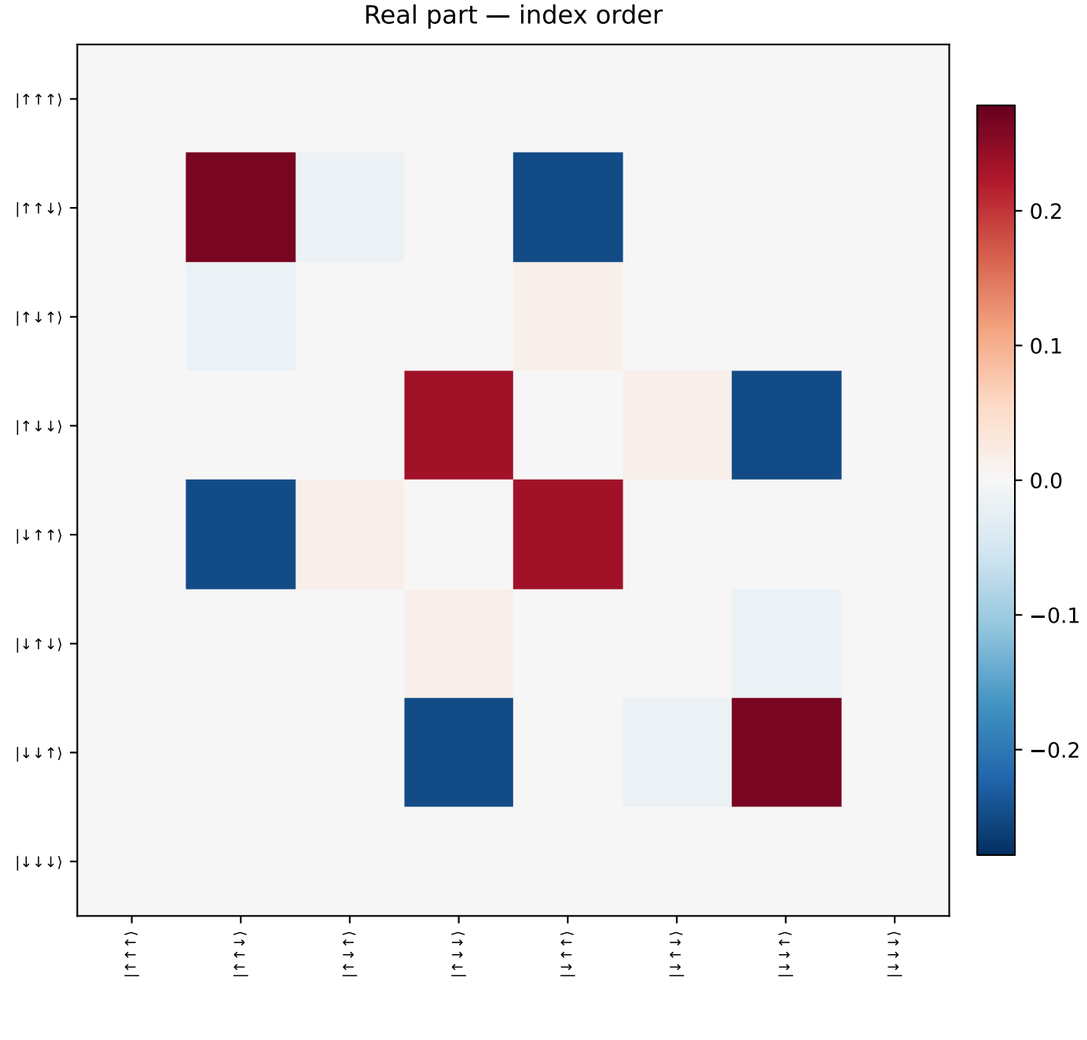
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.44523254

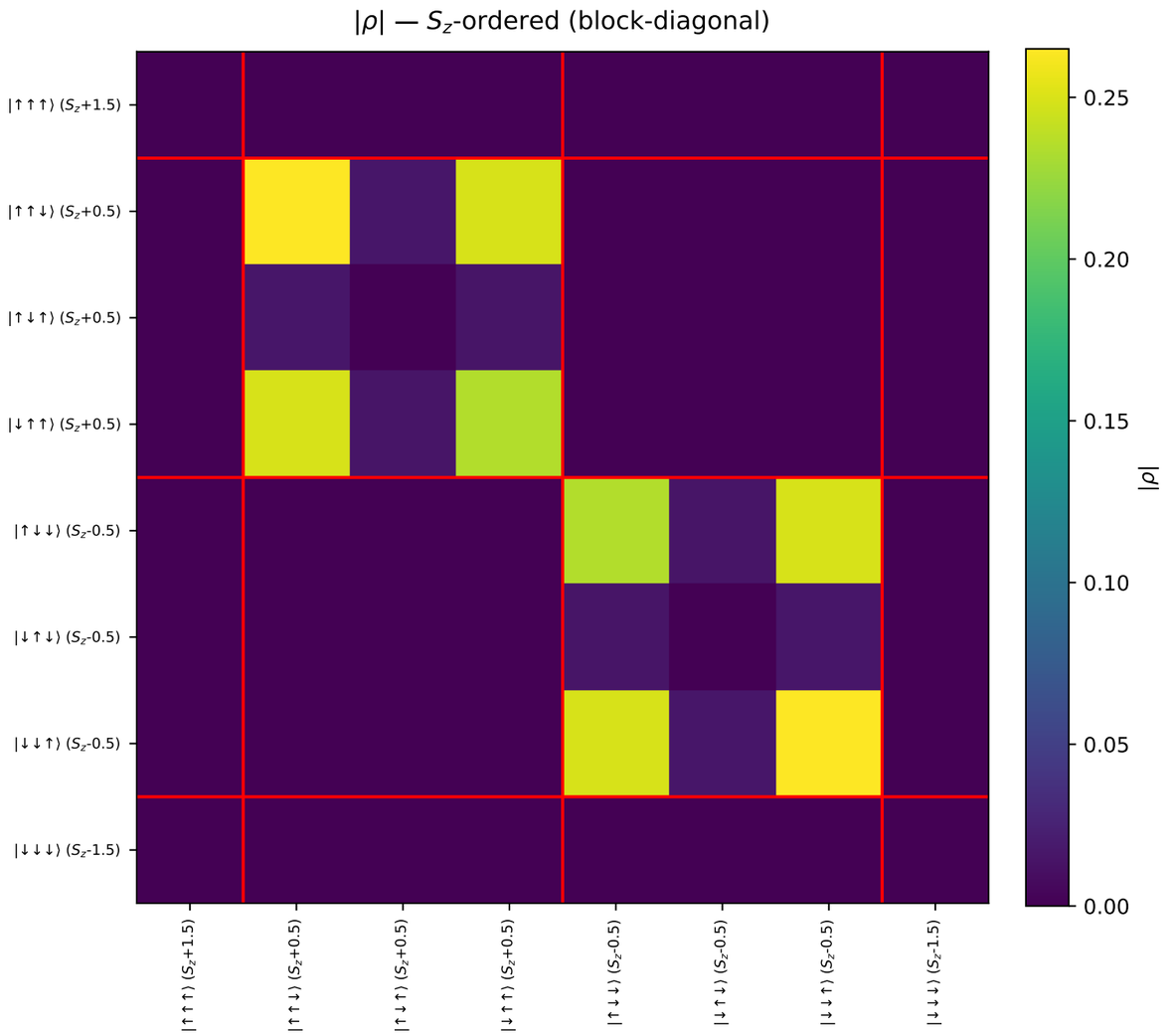
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.111953  C_GME=0.000000
2. p=0.081882  C_GME=0.000166
3. p=0.104404  C_GME=0.000000
4. p=0.093107  C_GME=0.000000
5. p=0.115541  C_GME=0.000000
... and 5 more states
```

3-Body RDM (Sites 2,3,4)





Trace=1.000000 Purity=0.499782 Entropy=0.695444 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 2,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.4997819190
Von Neumann entropy	0.6954436893
Eigenvalue min	3.635211e-05
Eigenvalue max	4.998909e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.49978192

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis
==========

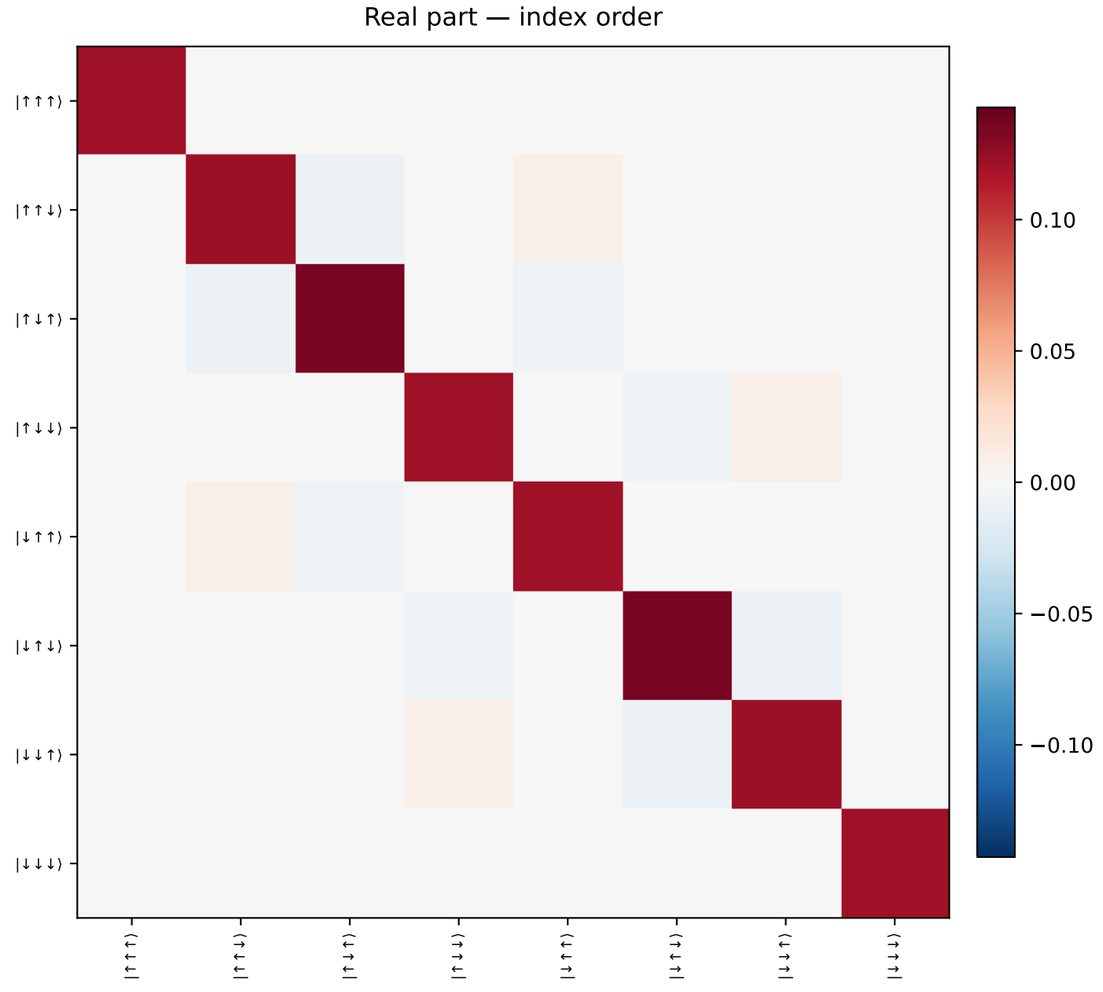
Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.08790080

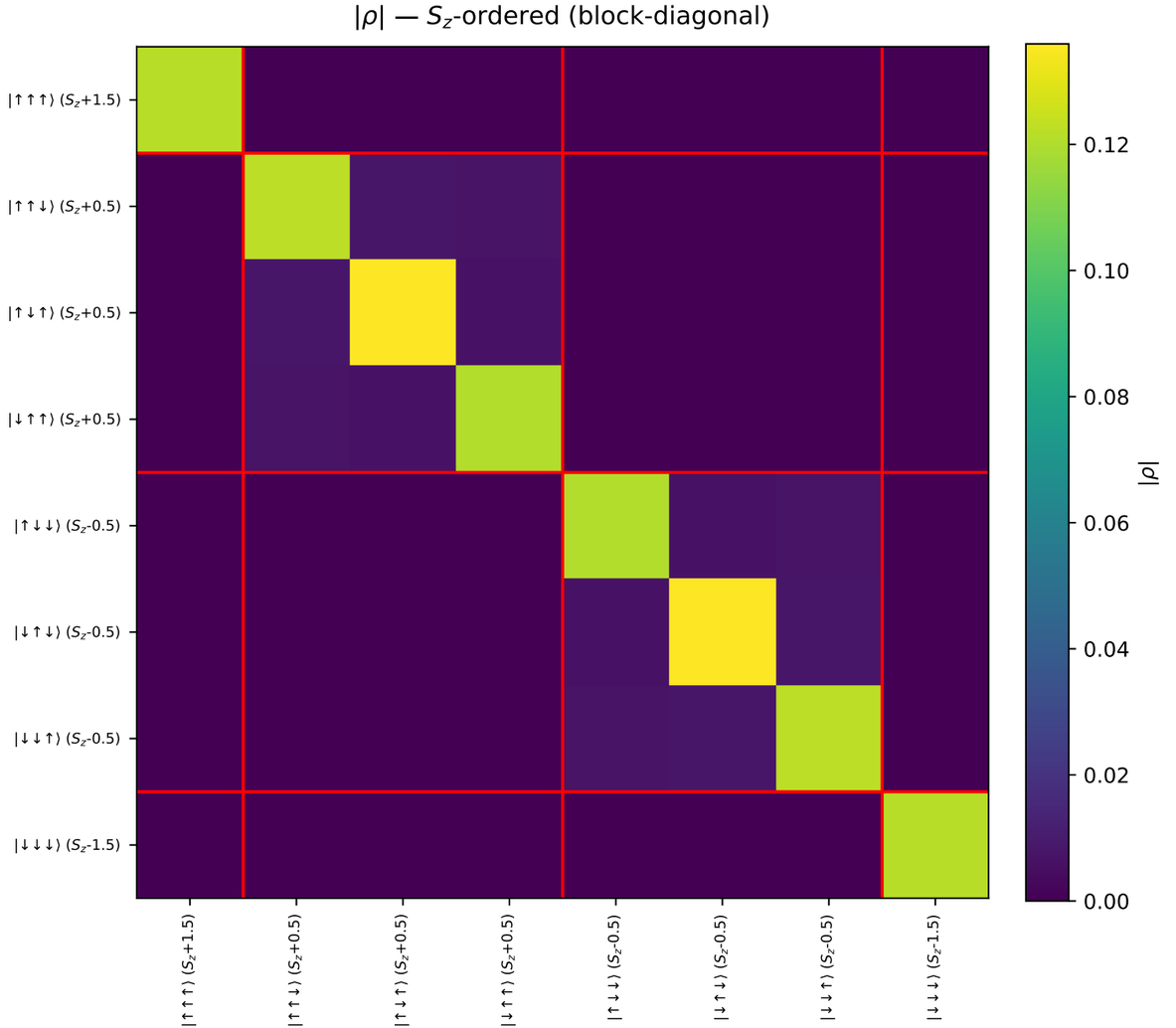
Status: inconclusive

Solver source:
–

1. p=0.077701 C_GME=0.077348
2. p=0.069934 C_GME=0.045648
3. p=0.098935 C_GME=0.055351
4. p=0.163311 C_GME=0.070531
5. p=0.076923 C_GME=0.099885
... and 4 more states

3-Body RDM (Sites 2,3,5)





Trace=1.000000 Purity=0.125967 Entropy=2.075664 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 2,3,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1259673001
  Von Neumann entropy  : 2.0756636393
  Eigenvalue min       : 1.140523e-01
  Eigenvalue max       : 1.433427e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.12596730

  A            | B            | Tr(rho_A^2)
  ─────────────| ─────────────| ──────────

STATUS: MIXED STATE
──────────────────────────────────

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

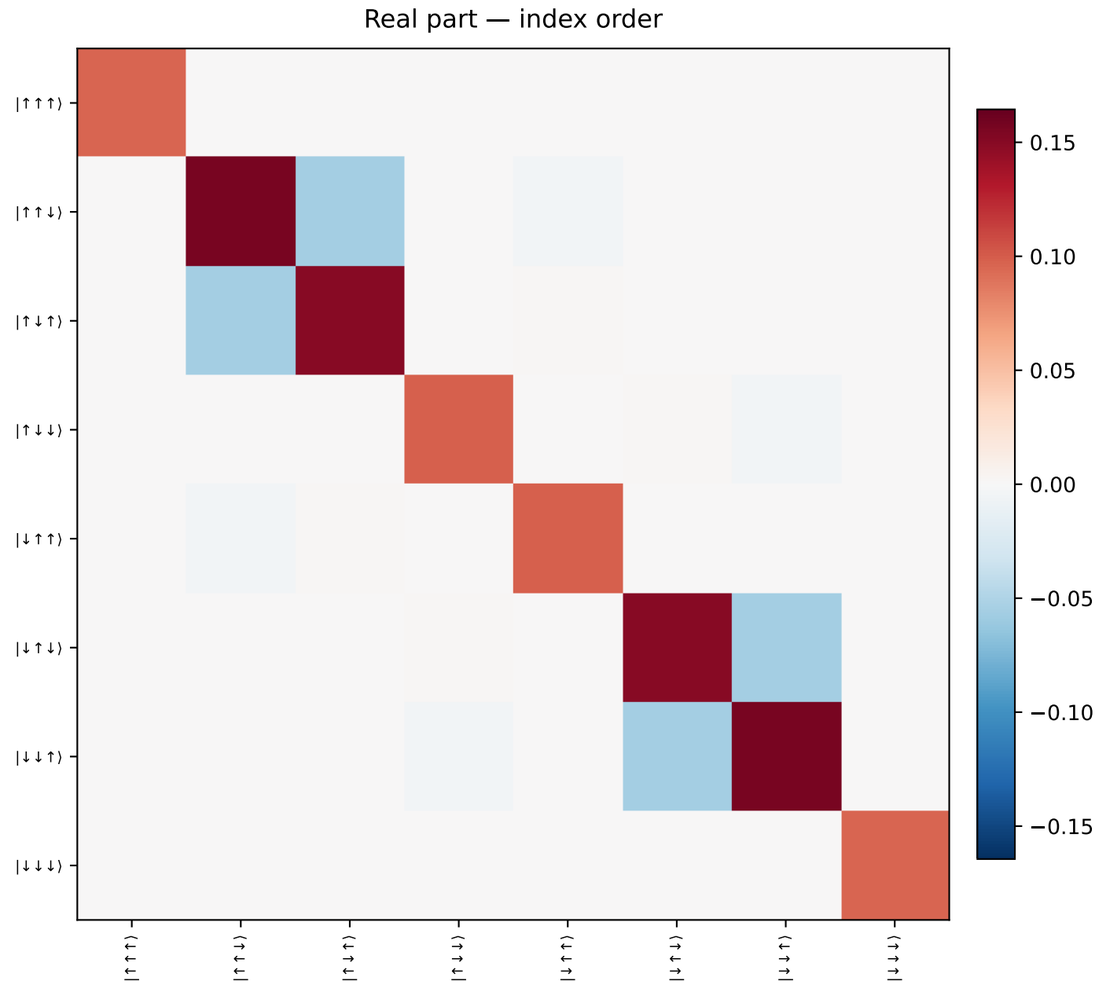
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.44564732

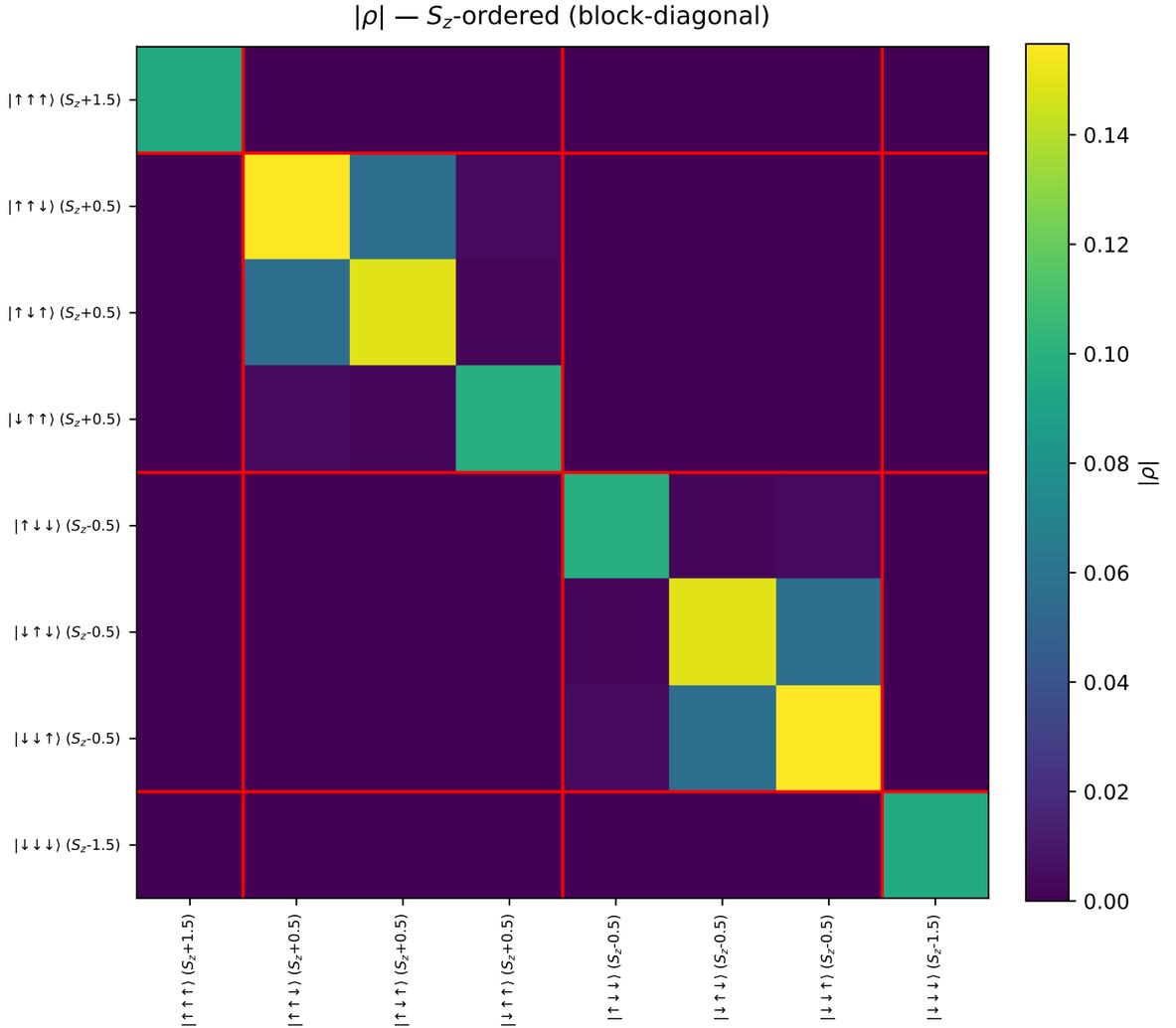
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.111700  C_GME=0.001424
2. p=0.080110  C_GME=0.000000
3. p=0.091762  C_GME=0.000000
4. p=0.093797  C_GME=0.000001
5. p=0.113797  C_GME=0.000491
... and 5 more states
```

3-Body RDM (Sites 2,4,5)





Trace=1.000000 Purity=0.144255 Entropy=2.010634 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 2,4,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1442552519
  Von Neumann entropy  : 2.0106344236
  Eigenvalue min       : 9.531950e-02
  Eigenvalue max       : 2.099249e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.14425525

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
===================================

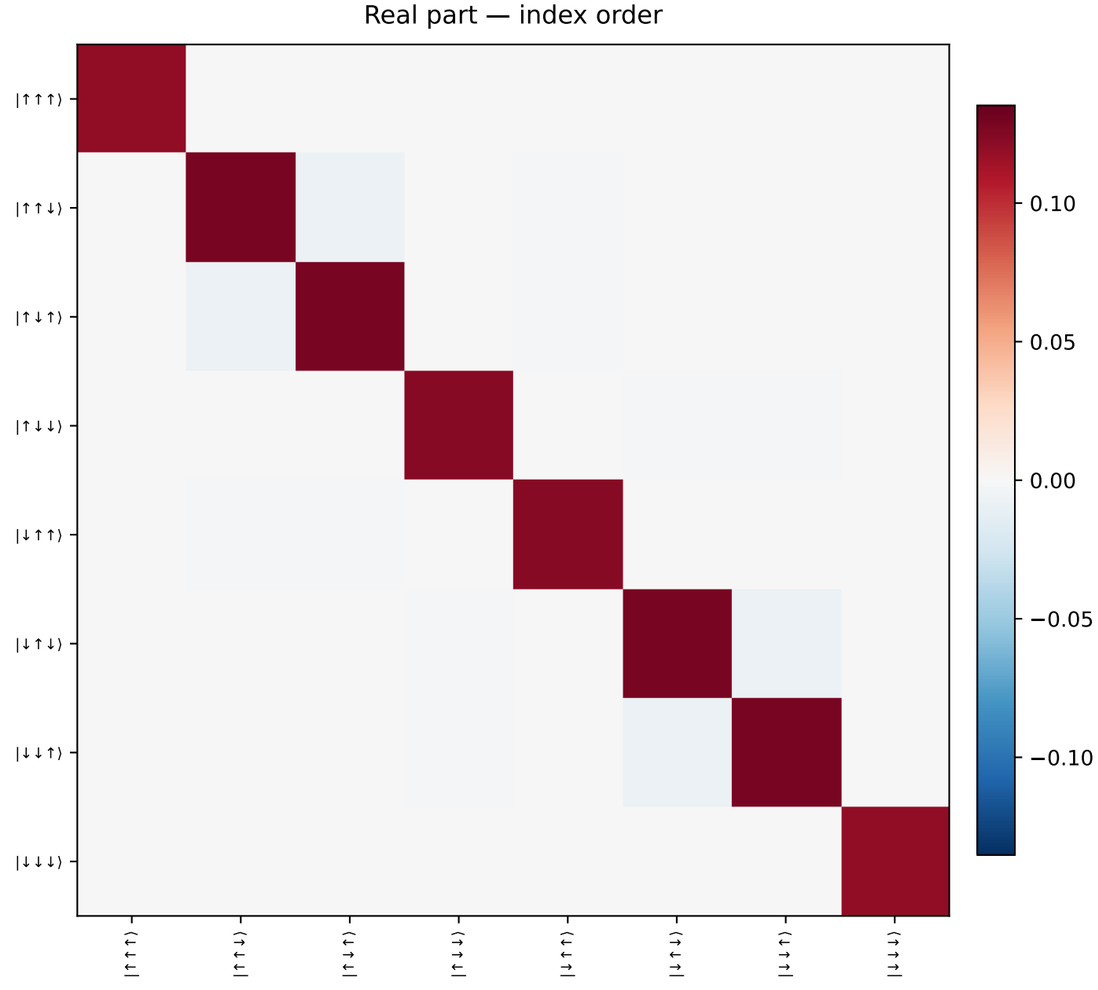
  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.35302550

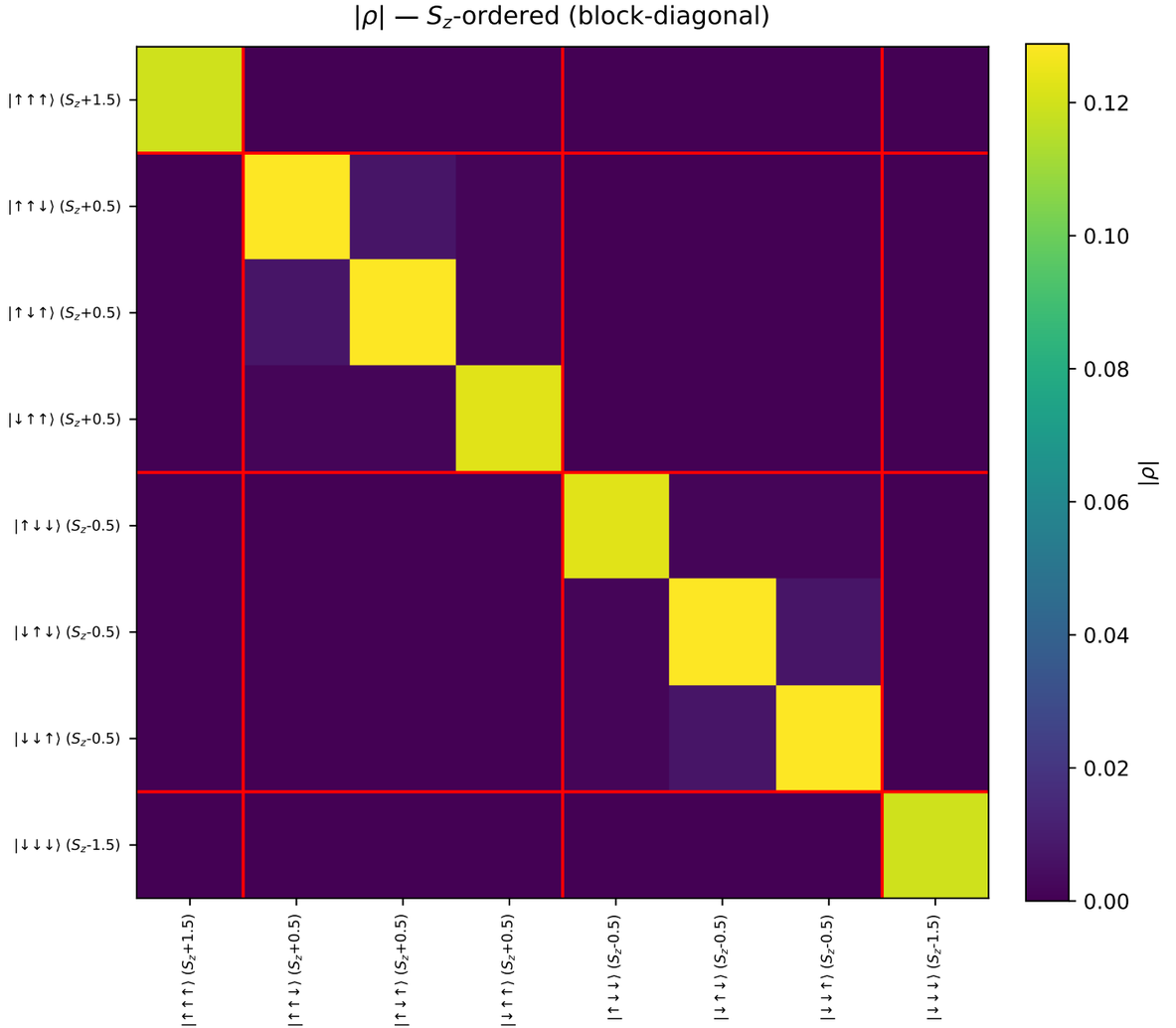
  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.067451  C_GME=0.000006
2. p=0.075601  C_GME=0.000003
3. p=0.142542  C_GME=0.000001
4. p=0.116862  C_GME=0.000007
5. p=0.038147  C_GME=0.000004
... and 5 more states
```

3-Body RDM (Sites 3,4,5)





Trace=1.000000 Purity=0.125368 Entropy=2.077991 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 3,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.1253680220
Von Neumann entropy	2.0779912025
Eigenvalue min	1.195895e-01
Eigenvalue max	1.361948e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.12536802

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex Roof Analysis
==========

Lower Bounds:
  C_LB = 0.00000000
C_GME (unitary optimization):
  C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.41380586

Status: no_GME

Solver source:
  –

1. p=0.094261 C_GME=0.000001
2. p=0.078142 C_GME=0.000031
3. p=0.097092 C_GME=0.000026
4. p=0.110634 C_GME=0.000001
5. p=0.110665 C_GME=0.000000
... and 5 more states